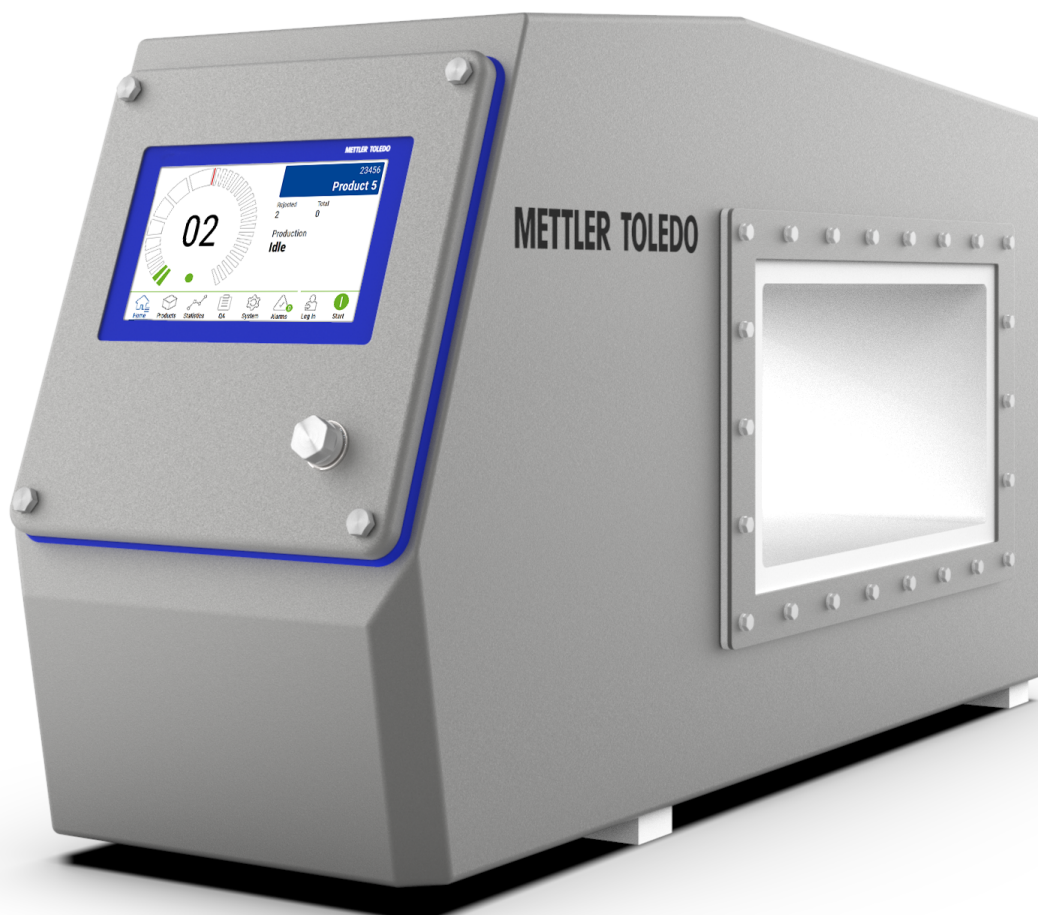


Metal Detector

M31R / M33R / M33RB / M34R



METTLER TOLEDO

Table of Contents

1	Introduction	7
1.1	Original Language	7
1.2	Additional Documents	7
1.3	Contacting METTLER TOLEDO Service	7
1.4	Manufacturer Maintenance	7
1.5	Warranty	7
2	Important Safety Information	8
2.1	Intended Audience	8
2.2	Hazard Notifications	8
2.2.1	Definition of Hazard Severity Panels	8
2.3	Intended Use and Foreseeable Misuse	9
3	Important Notice	10
3.1	Non-Ionizing Radiation	10
3.2	Essential Setup Tasks after Installation	10
3.3	Firmware Versions	11
3.4	METTLER TOLEDO End User License Agreement (EULA) for Software	11
4	Strategy and Principles	12
4.1	Your Metal Detection Strategy	12
4.1.1	The METTLER TOLEDO Metal Detection Guide	12
4.2	General Considerations when Optimizing Metal Detection	12
4.2.1	Your Products for Inspection	12
4.2.2	Understanding Vibration Signals	13
4.2.3	Understanding Product Effect	13
4.2.4	Understanding Phase and Amplitude	13
4.2.4.1	Phase Control	13
4.2.5	Understanding Choice of Operating Mode	14
4.2.6	Understanding Programmable Reject Timers	14
4.2.7	Understanding Speed	14
4.2.8	Understanding Reject Confirmation	14
4.2.9	Differences between StandardLine and PlusLine Metal Detectors	14
5	Description of the User Interface (UI)	16
5.1	Pages	16
5.1.1	Home Page	16
5.1.1.1	Changing the Home Page View	18
5.1.2	Products Pages	18
5.1.2.1	Changeover Page	18
5.1.2.2	Auto Setup Page	19
5.1.2.3	Setup Wizard Pages	19
5.1.2.4	Product List Page	19
5.1.2.5	Product Settings Pages	20
5.1.2.6	Product Rules Page	20
5.1.3	Statistics Pages	21
5.1.3.1	Production Page	21
5.1.3.2	Settings Page	21
5.1.3.3	Detection Log	21
5.1.4	QA Pages	21
5.1.4.1	Performance Test Page	21
5.1.4.1.1	Reminder Only Test	22
5.1.4.1.2	Standard Test	22
5.1.4.2	Test Setup Pages	22
5.1.4.3	Test Sample Setup Page	23
5.1.4.4	History Page	23
5.1.4.5	Reports	24
5.1.4.6	Settings	24
5.1.4.7	Access Log	24

5.1.5	System Pages	25
5.1.5.1	Information Pages.....	25
5.1.5.2	Region Settings Pages.....	25
5.1.5.3	Users Pages.....	26
5.1.5.3.1	Accounts Page	26
5.1.5.3.2	My Account Pages	26
5.1.5.3.3	Access Log	27
5.1.5.3.4	Access Log Settings	27
5.1.5.4	Auto Setup Settings Pages.....	27
5.1.5.5	Speed Settings Page.....	27
5.1.5.6	System Integration Pages.....	28
5.1.5.6.1	Inputs Page	28
5.1.5.6.2	Outputs Page	28
5.1.5.6.3	Conveyor Page.....	28
5.1.5.6.4	PackML Page.....	29
5.1.5.7	Detection and Rejection Page	29
5.1.5.8	Communications Page	29
5.1.5.9	ProdX Settings Page.....	29
5.1.5.10	OPC UA	30
5.1.5.11	Licensed Options Pages	30
5.1.5.12	Display and Buzzer Page.....	30
5.1.5.13	Alarm Settings Page	30
5.1.5.14	Alarm Log	31
5.1.5.15	Diagnostics Page.....	31
5.1.5.16	Self Service Page	31
5.1.5.17	Identification Page	31
5.1.6	Alarms Page	31
5.2	Keyboards and Character Sets	32
5.3	Nudge Editors	32
6	All Users: Managing your Account	33
6.1	User Accounts and Roles.....	33
6.2	Default Accounts and Passwords	33
6.3	Logging In.....	33
6.3.1	Automatic Logout.....	33
6.4	Changing your Account Settings.....	34
7	Administrators: Managing User Accounts	35
7.1	Creating an Account	35
7.2	Renaming an Account.....	35
7.3	Changing Passwords.....	36
7.4	Changing the UI Language for a Specific Account.....	36
7.5	Changing the Account Role.....	36
7.6	Enabling and Disabling an Account.....	37
7.7	Managing Performance Test Access	37
7.8	Deleting an Account.....	37
8	Setting System Language, Date and Time	38
8.1	Setting the System UI Language	38
8.2	Setting the System Date and Time	38
9	System Integration	39
9.1	Programmable Inputs and Outputs	39
9.1.1	Programmable Input List.....	39
9.1.2	Programmable Output List	39
9.1.3	Input and Output Configuration	40
9.1.3.1	Input Configuration.....	40
9.1.3.2	Output Configuration	41
9.2	Managing Inputs	41
9.2.1	Setting up Inputs	41
9.2.2	Unused	42
9.2.3	Acknowledge All Alarms Input.....	42

9.2.4	Acknowledge And Reset All Alarms Input.....	42
9.2.5	Air Supply Sensor Input.....	43
9.2.6	Bin Full Sensor Input.....	43
9.2.7	Bin Unlocked Input.....	44
9.2.8	Conveyor Running Input.....	44
9.2.9	Conveyor Start Input.....	44
9.2.10	Conveyor Stop Input.....	44
9.2.11	Downstream Blocked Input.....	45
9.2.12	Pack Sensor Input.....	45
9.2.13	Reject Check Sensor Input.....	45
9.2.14	Reject Operation Sensor Input.....	45
9.2.15	Rejected Product Sensor Input.....	46
9.2.16	Reset All Alarms Input.....	46
9.2.17	Safety Circuit Input.....	47
9.2.18	Speed Sensor Input.....	47
9.2.19	Upstream Starved Input.....	47
9.3	Managing Outputs.....	48
9.3.1	Setting up Outputs.....	48
9.3.2	Unused.....	48
9.3.3	Air Supply Failure Output.....	48
9.3.4	Bin Full Output.....	48
9.3.5	Bin Unlock Output.....	49
9.3.6	Bin Unlocked Output.....	49
9.3.7	Conveyor Drive Output.....	49
9.3.8	Current Alarms Output.....	49
9.3.9	Current Fault Alarms and Higher Output.....	50
9.3.10	Current Warning Alarms Output.....	50
9.3.11	Current Warning Alarms and Higher Output.....	50
9.3.12	Detect Output.....	50
9.3.13	Good Pack/Product Output.....	51
9.3.14	Latched Alarms Output.....	51
9.3.15	Performance Test Due Output.....	51
9.3.16	Performance Test Overdue Output.....	51
9.3.17	ProdX Disconnected Output.....	51
9.3.18	Reject Output.....	51
9.3.19	Reject Not Confirmed Output.....	52
9.3.20	Stop On Detection Output.....	52
9.3.21	System Healthy Output.....	52
9.3.22	System Stop Output.....	53
9.3.23	Unacknowledged Alarms Output.....	54
9.4	Speed Settings.....	54
9.4.1	Speed Sensor.....	54
9.4.2	Setting up Conveyor Speed.....	54
9.5	Conveyor Start/Stop Control Options.....	55
9.5.1	The Start/Stop Control.....	55
9.6	Configure the Pack Sensor.....	56
9.7	Machine Status (PackML State).....	56
9.8	Bin Unlock Function.....	56
9.8.1	Unlocking the Bin.....	56
9.8.2	Bin Unlocked Input.....	57
9.8.3	Bin Unlock Outputs.....	58
9.8.3.1	Bin Unlock.....	58
9.8.3.2	Bin Unlocked.....	58
9.9	ProdX.....	59
9.9.1	Manual ProdX Setup.....	59
9.10	Reject Check Sensor.....	59
9.10.1	Set-up the Reject Check Sensor.....	60
9.10.1.1	Reject Check Sensor Main I/O Page Description.....	60
9.10.1.2	Reject Check Sensor Main I/O Page Set-up.....	60
9.10.2	Reject Check Error Messages.....	61

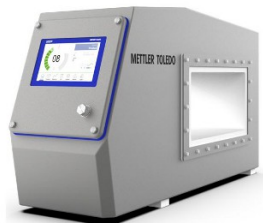
10	Managing Products	62
10.1	Preconditions / Considerations.....	62
10.2	Setting up Products	62
10.2.1	Setting up a Product for the First Time	63
10.2.1.1	Product Setup.....	63
10.2.1.2	Adjust Inspection Settings	63
10.2.1.3	Synchronized Reject Setup	63
10.2.1.4	Final Adjustments	64
10.2.2	Creating a New Product from an Existing Product	64
10.3	Viewing and Changing Products	64
10.4	Editing Products and Product Settings.....	65
10.4.1	Set Mode Selection Manually	65
10.4.2	Adjust Sensitivity and Phase Manually.....	66
10.4.3	Set Signal Level for Auto Setup.....	66
10.5	Manually Adjust Non-Synchronized Reject Settings.....	66
10.6	Manually Adjust Synchronized Reject Settings.....	67
10.7	Viewing and Resetting the Counters and Settings.....	68
11	Performance Testing	70
11.1	Test Sample Setup.....	70
11.2	Setting up Performance Tests.....	70
11.2.1	StandardLine	70
11.2.2	PlusLine	71
11.3	Carrying out a Performance Test	71
11.3.1	Performance Test - Reminder Only Test	72
11.3.2	Standard Test	72
12	Event Logs and Reports	74
12.1	Filtering Logs and Reports	74
12.1.1	Example - QA > History	74
12.2	Detection Log.....	75
12.2.1	Downloading Detection Logs.....	75
12.2.2	Deleting the Detection Log.....	76
12.3	History	76
12.3.1	Downloading History Logs	77
12.3.2	Deleting the History Log	77
12.4	Access Log.....	77
12.4.1	Downloading User Access Logs.....	77
12.4.2	Deleting Access Logs	78
12.5	Alarms Log.....	78
12.5.1	Downloading Alarm Logs.....	79
12.5.2	Deleting Alarm Logs	79
12.6	Performance Test	80
12.6.1	Downloading the Performance Test.....	80
12.6.2	Deleting the Performance Test Log	80
13	System Updates	82
13.1	OS Update.....	82
13.2	System Update.....	82
13.3	IO Expanders	83
14	Managing Alarms	84
14.1	Alarm Icons and their Meaning	84
14.2	Choosing Alarm Settings	84
14.2.1	Choosing the Display Priority	85
14.2.2	Choosing how Alarms are Acknowledged	85
14.2.3	Choosing how Alarms are Reset	85
14.3	Acknowledging Alarms.....	86
14.4	List of Alarms.....	86

15	Preventative Maintenance	91
16	Technical Specifications	92
16.1	Metal Detectors	92
16.2	Feature Packs	92

1 Introduction

This manual covers the following M30 R-series metal detectors:

- M31R
- M33R
- M33RB
- M34R



M30 R-Series Metal Detector

1.1 Original Language

The original manual is written in English. If you are reading a translated version of this manual, and you also need the English original manual, you can ask METTLER TOLEDO to supply it. If you have a question about the intended meaning of any translated text, consult the original English-language manual.

1.2 Additional Documents

The following additional documents may be included with your equipment:

- Installation Manual
- Declaration of Conformity
- Guide to Metal Detection Technology

1.3 Contacting METTLER TOLEDO Service

Contact METTLER TOLEDO Service, using the link ► <http://www.mt.com/contact>, about the following products and services:

When you contact METTLER TOLEDO Service, have the following information available, if applicable:

- METTLER TOLEDO order number and date
- Equipment name, model, or type
- Serial number
- Production line name
- Pictures or videos of the part or problem
- Firmware version
- Precise wording of the displayed alarm message with fault code

1.4 Manufacturer Maintenance

In case of any worn or damaged component, contact METTLER TOLEDO Service.

1.5 Warranty

For information on warranty, refer to the official METTLER TOLEDO terms and conditions at:

► http://www.mt.com/us/en/home/site_content/legal/commercial_terms.html.

2 Important Safety Information



WARNING

Metal detectors, and metal detector conveyor systems, have several hazards.

Familiarize yourself with the safety information in the conveyor system Operator Manual and Installation Manual before you operate the metal detector.

The metal detector head has no moving parts. When integrated with a METTLER TOLEDO conveyor system, additional hazards and risks, such as suddenly moving parts, have been mitigated.

Where a client has integrated a head into their own conveyor system or operates it in any other way, it is their responsibility to ensure adequate operating safety measures are in place and are adhered to.

2.1 Intended Audience

This Operator Manual contains information and instructions for operating the metal detector, including system and product settings. It may also serve as a reference for METTLER TOLEDO service technicians.

An operator is a person who performs acts that are routine, repetitive, and integral to the use of the equipment for production, and has access only to areas which do not require tools, or higher security levels, where routine, repetitive and integral are defined as follows: Routine - performed as part of a regular, basic course of procedure; Repetitive - repeated regularly as part of the production process or cycle; and Integral - inherent to, and be performed as part of, the production process.

A technical person is a person qualified with higher levels of training, reasoning, judgment, responsibility, and authority, who performs tasks that are not necessarily routine, and has access to areas requiring tools or higher security levels. A qualified person is a person who, by possession of a recognized degree or certificate of professional standing or by extensive knowledge, training, and experience, has successfully demonstrated their ability to solve problems relating to the subject matter and work.

2.2 Hazard Notifications

A hazard notification consists of the following:

- Hazard alert symbol (yellow triangle with black symbol)
- Signal word (**DANGER**, **WARNING**, or **CAUTION**)
- Special notifications related to the hazard (as required)

The hazard severity panels are attached next to the labels for the hazard alert symbols on the equipment.

2.2.1 Definition of Hazard Severity Panels

Hazard severity panels describe the level of risk of a particular hazard. The background color shows the risk as shown in the following table. According to ISO 3864, a hazard severity panel can be used as an alternative to a signal word. The labels on the equipment may contain both options. In this manual, however, hazard severity panels are always used together with their signal words. The following definitions are based upon the ISO 3864 definitions.

	(red): This hazard severity panel indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
	
	(orange): This hazard severity panel indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
	
	(yellow): This hazard severity panel indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.
	

2.3 Intended Use and Foreseeable Misuse



Important Safety Message

Read and understand all the safety information in the following sections as well as the safety messages in the rest of this manual.

If you do not follow the safety information and messages, this may lead to personal injury up to and including death as well as damage to the equipment and other property.

Only use the equipment as it was intended according to the procedures in this manual.

Any other use of the equipment is considered misuse. If you use the equipment beyond the limits of the design specification without written consent from METTLER TOLEDO, it is considered misuse.

Common foreseeable misuses include the following:

- Not following the instructions and important safety information described in this manual
- Using the equipment in a specific environment, unless the equipment is designed for that environment, for example:
 - Hygienic environments
 - Hazardous locations (explosive atmospheres)
 - Environments where there is need for aggressive cleaning or sanitization
- Placing anything on the equipment that is not designed to be placed on the equipment
- Not maintaining the equipment according to the instructions in the manual
- Using pens, tools, or any other small item instead of a finger on the HMI
- Doing tasks not authorized by your employer

3 Important Notice

3.1 Non-Ionizing Radiation



⚠ WARNING

Risk of injury or death, electromagnetic fields inside and directly outside the aperture of metal detectors can affect the operation of cardiac pacemakers and other active or passive medical devices (implanted or external).

Personnel fitted with these types of devices must always consult their doctor or physician before they operate or come into close proximity of the metal detector aperture.



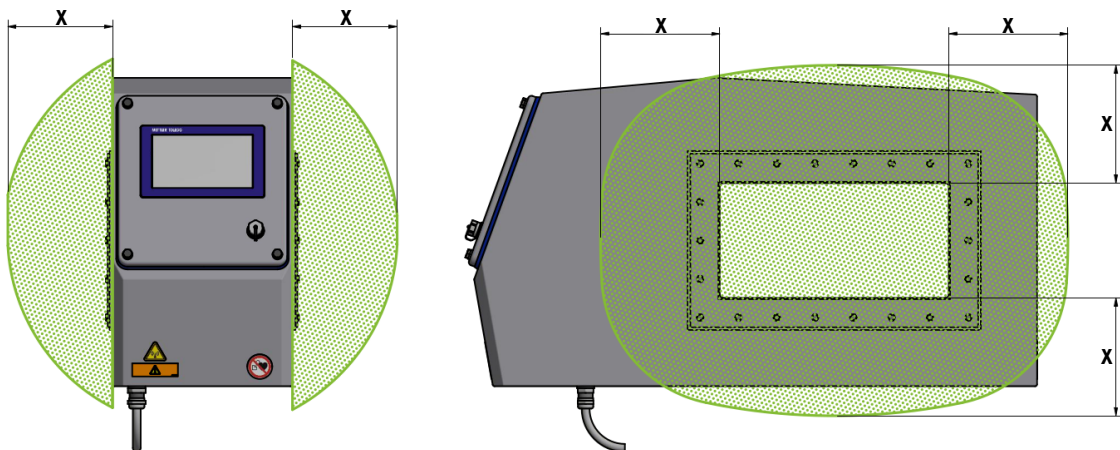
⚠ CAUTION

Non-ionizing radiation.

This is a radio-frequency metal detector that generates electromagnetic fields within the frequency range 10 kHz to 1 MHz. When the metal detector is operating, avoid prolonged exposure to the electromagnetic field inside the aperture and up to 200 mm from the aperture.

Inside the aperture of the detector the magnetic field strength is typically up to 600 A/m (RMS) across all operating frequencies, head types and aperture sizes.

At a distance of 200 mm directly outside of the aperture (X) the magnetic field strength is less than 1 A/m (RMS) across all frequencies, head types and apertures.



Non-ionizing radiation

The maximum electric field inside the aperture is typically 10 V/m (RMS) with negligible values immediately outside the aperture.

The current and energy induced in the human body resulting from exposure to the field inside the aperture comply with published basic restriction levels for occupational exposure.

If specific values are required per detector type, size and operating frequency then contact METTLER TOLEDO.

3.2 Essential Setup Tasks after Installation

METTLER TOLEDO encourage users of metal detectors to purchase the metal detector integrated with a METTLER TOLEDO conveyor/transport system with integrated reject mechanism. METTLER TOLEDO manufacture a standard range of systems, and non-standard custom systems for your factory to meet local requirements.

If you purchased the M30 R-Series metal detector integrated with a METTLER TOLEDO conveyor system, the settings, including the software definitions of the electrical inputs and outputs, should be correct for your requirements.

If you purchased an M30 R-Series metal detector to integrate with your own conveyor system and reject mechanism, you need to configure the system inputs/outputs via the System menu screens after you have completed the mechanical and electrical installation.

See below for further details.

See also

 System Integration ► Page 39

3.3 Firmware Versions

Some features within this manual may not be available with older versions of firmware.

To enquire about firmware upgrades, contact METTLER TOLEDO Service (See Contacting METTLER TOLEDO Service). METTLER TOLEDO Service will ask you for the Serial Number and the Firmware Version of the Metal Detector.

To view the Serial Number and the Firmware Version, navigate to System > Information.

3.4 METTLER TOLEDO End User License Agreement (EULA) for Software

The software on this product is subject to the METTLER TOLEDO End User License Agreement. Refer to ► <http://www.mt.com/legal>

4 Strategy and Principles

4.1 Your Metal Detection Strategy

The reliable detection of metal contamination in product-inspection environments presents complex scientific, engineering and data-processing challenges.

The standards for metal detection vary from company to company, and from nation to nation. The methods of manufacture of products such as food, drink and pharmaceuticals are constantly changing, even the ways that products are packaged can change suddenly (sometimes in response to changes in consumer taste or buying habits). A good metal detection strategy must account for all these challenges.

Research-and-development scientists, engineers and software developers at METTLER TOLEDO work as an agile team. We are dedicated to providing trusted solutions that meet your present challenges of metal detection, and to responding quickly to any new challenges you might encounter.

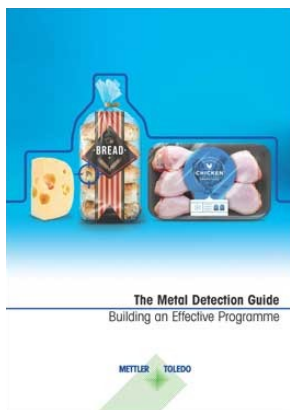
To get the most from the equipment software, read the METTLER TOLEDO Metal Detection Guide. The guide explains our metal detection technology in detail. The guide contains advice to help you to build the most effective metal detection strategy for your production line, including advice on the position of metal detection systems, from the raw-ingredient stage to packaged products.

METTLER TOLEDO Service can visit your site to advise on the best strategy, and to commission or re-commission the metal detector and its associated systems.

4.1.1 The METTLER TOLEDO Metal Detection Guide

METTLER TOLEDO recommends that system designers, installers, and quality assurance managers study the document called "Guide to Metal Detection Technology - Building an Effective Programme" that accompanies this manual.

The guide will help you to build the most effective metal detection program for your production line, and it will help you to get the most from the software features explained in this manual. The guide is updated periodically. You can download the most recent version, in your preferred language, for no charge, at [► Guide to Metal Detection Technology \(mt.com\)](#).



Also at [► Expertise Library - METTLER TOLEDO \(mt.com\)](#) you can download white papers and multimedia guides that can help your company's production-line designers and quality assurance managers.

4.2 General Considerations when Optimizing Metal Detection

The topics in this chapter explain important fundamentals of metal detection technology and how to get the most from your metal detection software. We advise you to read this chapter before you start to set up products to use the metal detector.

4.2.1 Your Products for Inspection

M30 R-Series metal detectors have an Auto Setup feature that allows you to easily set up the metal detector by simply passing the product for inspection through the aperture, and following the Auto Setup instructions on the screen. Every new product, and its settings, is memorized in the metal detector, and identified by the product name.

Auto Setup calculates the best Sensitivity, Phase, and Mode for the product and for the metal detection standards that you aim to achieve. Auto Setup works in conjunction with other metal detector settings and features. Advanced users, with a good knowledge of METTLER TOLEDO's metal detection technology, can make fine adjustments to the metal detector's settings.

Whether you use the Auto Setup alone or Auto Setup followed by advanced manual setup, it is beneficial to understand the principles of metal detection. The following sections of the manual explain what we mean by phase, phase angle, amplitude, sensitivity, reject mechanisms, and operating frequency considerations.

4.2.2 Understanding Vibration Signals

M30 R-Series metal detectors are designed to minimize the effects of vibration, such as the risk of unwanted signals that could cause a false reject. However, some vibration, including from the product-conveyor system, is inevitable.

The signals generated from vibration of the coil system in the metal detector have a characteristic phase angle. The phase angle of the vibration signal is used as a reference. The phase angle of any metal contamination is measured relative to the phase angle created by vibration effects.

4.2.3 Understanding Product Effect

Dry products are easier to inspect for metal contamination than wet products. This is because wet products, and products with high salt and moisture content, are conductive. Wet products include fresh meat, poultry, cheese, fruit, vegetables, beverages and fish. They create a "product effect" signal which must be distinguished from the signal created by a metal contaminant. Similarly, certain types of product packaging, such as metallized film, can create a product effect.

Vibration and product effect signals can easily be reduced by adjusting the sensitivity control, however, reducing the sensitivity affects all signals: metallic signals, vibration signals, and product signals. Phase Control is used to discriminate between wanted and unwanted signals.

For a good understanding of product effect, and the challenges it presents, refer to the [► The METTLER TOLEDO Metal Detection Guide](#). In particular, refer to the sections "Factors Limiting Sensitivity" and "Understanding Challenging Applications for Improved Metal Detection".

4.2.4 Understanding Phase and Amplitude

M30 R-Series metal detectors use low-power radio-frequency transmit and receive coils. This system can sense tiny disturbances created by metal particles passing through the aperture of the metal detector. The signal generated is characterized by a phase angle and amplitude.

A metal particle passing through the aperture of the detector changes the **phase angle** and **amplitude** expected in the receiving coils.

Phase Angle

Different metals, and different types of product for inspection, produce signals that have a phase angle.

When the metal detector is subject to mechanical vibration, the vibration causes a known phase angle. Any phase angle caused by product passing through the metal detector is referenced against the phase angle known to be caused by vibration.

Amplitude

The amplitude is the size of the signal received by the metal detector. In general, the larger the metal contaminant, the larger the signal amplitude.

Although a metal particle will create a signal with phase and amplitude, this signal needs to be computed and referenced against the phase and amplitude expected from the type of product under inspection. The signals from dry products, whether loose or in packs, are easier to distinguish from the signal created by a metal particle. The signals from wet and salty products, and products packaged in metallized film, are more challenging to distinguish from the signals from metal contamination.

4.2.4.1 Phase Control

Radio-frequency metal detectors determine any changes in the phase angle of the signal generated by the receiver coils. Therefore, the signal caused by vibration, and the active product, must be minimized.

The phase control circuitry discriminates between the wanted signals from metal particles and the unwanted signals from vibration and product effect.

4.2.5 Understanding Choice of Operating Mode

M30 R-Series metal detectors are supplied with several Mode configurations that depend on the type of equipment and options that you purchased. Some models have multiple simultaneous frequency modes, that makes the metal detector better able to detect contamination of wet products, by discerning and cancelling out the product signal. The Mode selection determines the frequency. The Mode is in the Products pages. It is setup individually for each product in the Product List.

METTLER TOLEDO advise that you use the Auto Setup wizard routine, where available, to set up each product. Auto Setup calculates the optimum frequency automatically. As a general rule, lower frequencies are better if the product under inspection is wet, and higher frequencies are better if the product under inspection is dry. The METTLER TOLEDO Metal Detection Guide explains the importance of frequency, and how frequency relates to dry products and wet products (and other products that create a "product effect").

4.2.6 Understanding Programmable Reject Timers

When metal is detected, either the conveyor system is stopped (Stop-on-Detect) or the contaminated product is automatically rejected by a reject mechanism. The mechanism activated to reject the contaminated product varies from system to system. Examples of reject mechanism are: punch, pusher, air blast, flap, diverter arm, and retracting belt.

For most products, the activation of the reject mechanism requires precise timing. The timing must take into account factors such as the conveyor speed (on conveyor systems), the size and spacing of packs (for discretely packaged products), the height and speed of a falling product (for metal detection in gravity fall applications). The speed of the reject mechanism, reject relay and reject solenoid must be considered, especially for products passing through the metal detector at high throughput.

Programmable reject timers in the metal detector electronics control all the different types of reject mechanism. The timing can be:

- **Immediate.**
- **Delayed:** this is necessary on system conveyors where the reject mechanism is some distance away from the metal detector. For delayed reject, the system conveyor can be fitted with a speed sensor.
- **Synchronized:** for products in discrete packs of a known size, the timing of the reject mechanism can be synchronized. For synchronized reject, the system conveyor must be fitted with a pack sensor. To simplify the setting up of synchronized reject mechanisms, and to ensure accurate reject operation, M30 R-Series metal detectors have a semi-automated timer setup routine.

4.2.7 Understanding Speed

The metal detector must know the speed of the conveyor. If the conveyor speed is fixed, the speed can be entered in the Speed Settings menu. If the conveyor speed is variable, the system must have a speed sensor fitted to inform the metal detector and to make sure that contaminated products are reliably rejected from the conveyor.

The metal detector operates with a wide speed range of the conveyor.

Note The speed can affect the phase and amplitude of the signal. Therefore fixed speed configurations are preferable.

4.2.8 Understanding Reject Confirmation

When a contaminant is detected, M30 Series metal detectors can integrate in a number of ways with the conveyor system, alarms, reject mechanisms, and reject confirmation.

Connectors on the User Input/Output board of the metal detector can be linked to a Reject Confirmation circuit on the conveyor system.

Reject confirmation provides feedback to the metal detector to confirm that the reject mechanism has operated correctly, and at the right time, to reject contaminated product. The metal detector can then give feedback to stop the conveyor or warn the operator.

4.2.9 Differences between StandardLine and PlusLine Metal Detectors

Key differences between StandardLine and PlusLine include available operating frequencies and modes for different end user applications.

M31R StandardLine is the entry level model, powered by Sense software, giving good overall, reliable and consistent performance suitable for most product types and applications.

SENSE™ is the system software that connects all the metal detection system elements, for real-time decisions and centralized control.

M33R, M33RB and M34R PlusLine detectors have expanded functionality (functionality packs fitted as standard), with potential for greater sensitivity performance in targeted applications such as dry, bulk and wet products.

PlusLine detectors also contain more user accounts (50, compared to 10 user accounts available to StandardLine models) and product memories (200, compared to 100 available to StandardLine models).

All models share a common user interface (UI). The content of the UI is dependant on the specification of the metal detector. Some screens may not be present, or the order in which they are accessed may differ. To identify the model of your metal detector, from the Home page:

- 1 Navigate to **System > Information > Basic**.
 - 2 The **Basic** page displays the **Model Type**.
- (You do not need to log in with specific access rights).

Any differences between models, where applicable, will be referenced at the beginning of each relevant topic as in the following example:

Metal Detector	Access Level Required
PlusLine - M34R	Engineer

5 Description of the User Interface (UI)

The metal detector is operated through a touchscreen user interface (UI). To operate the UI, use your fingers to tap the interactive elements shown on the screen.



NOTICE

Do not use sharp objects to operate the touchscreen.

Tap the screen with your fingers, with or without gloves.

The content of the UI is dependant on the specification of the metal detector. Some screens may not be present, or the order in which they are accessed may differ.

5.1 Pages

UI pages are organized in the following groups:

- Home
- Products
- Statistics
- QA
- System
- Alarms

This section describes all pages contained within each of the above groups.

Note The features discussed in the UI may appear on different pages depending on the system and feature packs installed.

The use of chevrons:

- Navigate to System > Information > Basic.
- Navigate to System > Information > > OS.
 - Single chevrons indicate direct path to the Basic page.
 - Double chevrons indicate indirect path to the Operating System (OS) feature, this is located on another Page (X of Z).

5.1.1 Home Page

Metal Detector	Access Level Required
All Metal Detectors	All Users

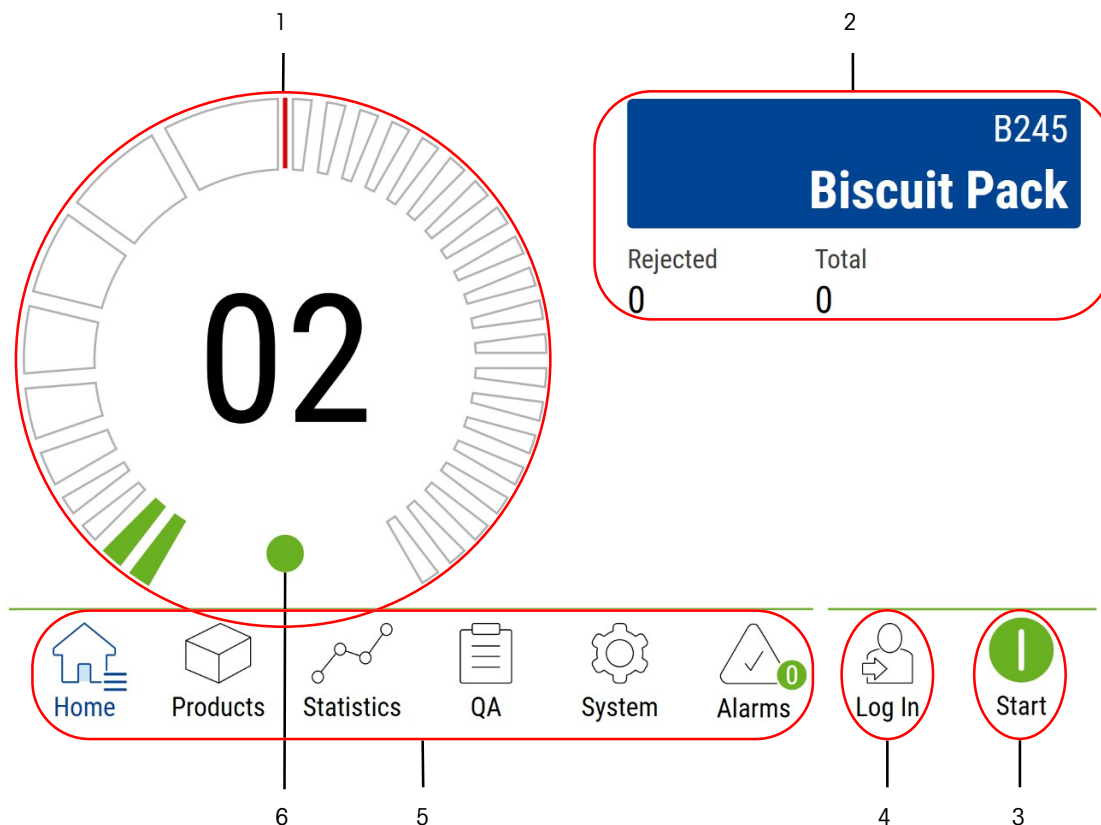
The Home page opens when the metal detector is started. From here you can access the rest of the UI.

The Home page requires no login. However, you may be asked to log in as you open other pages.

To learn more, see:

- [User Accounts and Roles ► Page 33]

The main elements of the Home page are as follows:



1	The Bargraph shows product signal strength. Moving past 12 o'clock, the bars become red and indicate a detection.
2	The Active Product area shows the Name and Code of the Active product. Tap anywhere inside the area to navigate to the Changeover page, where you can change the Active product.
3	The Start / Stop button starts and stops the conveyor. The button changes depending on the conveyor state: Start appears if the conveyor is stationary, Stop appears if the conveyor is running. Note The button will function only if it is configured in the System menu. If the UI is not configured to control the conveyor, the Start / Stop button is not shown. If the button appears grayed out, underlying conditions or alarms have indicated that the conveyor should not be started.
4	The Log In / Log Out button takes you to the Select account page, where you enter your login details. If you are already logged in, your account is shown above this button.
5	Each button on the Navigation Bar opens a section of the UI containing various features and settings. Access to some features and settings is restricted depending on the logged in user's role.
6	The Reject Indicator at the foot of the bargraph displays the product inspection result: • Stays clear when no product present (for synchronized reject). • Constant green when good (no contamination) continuous products pass. • Flashes green when good (no contamination) discrete products pass. • Turns Red when the reject is active.

Note Depending on system configuration, other information can be displayed on the UI Home page, these include:

- Performance Monitoring Routine status.
- ProdX status.
- PackML mode and status.
- Bin Unlock status.

See also

- 🔗 User Accounts and Roles ► Page 33
- 🔗 Good Pack/Product Output ► Page 51

5.1.1.1 Changing the Home Page View

Metal Detector	Access Level Required
All Metal Detectors	All Users

The Home page allows you to switch between two different views:

- Large Bargraph

This view shows a large bargraph occupying the left side of the page.

- Alarms

This view looks different depending on the active alarms in the system:

- If none exist, the view is similar to the Alarms page. It shows the number of rejects and that there are zero active alarms.
- If active alarms are present in the system, one of them appears in this view. The alarm to be shown is chosen automatically based on Display Priority set in System > Alarm Settings. You can see the name, code, severity icon, acknowledged status and some guidance on how to reset the alarm.

The Large Bargraph view is shown by default unless there are active alarms in the system.

To change:

- 1 Navigate to the **Home** page.

➔ When the Home page is open, the Home button on the navigation bar contains three parallel bars:



- 2 Select the **Home** button.

➔ The **Home Menu** page opens, allowing you to choose between the two views.

- 3 Select **Alarms**.

Note Unless there is an active alarm in the system, the Large Bargraph view will be shown again the next time you open the Home page.

See also

🔗 Alarm Settings Page ► Page 30

5.1.2 Products Pages

The Products section provides access to product information and settings.

The contents of this section vary depending on the active product.

If the active product requires setup:

- Changeover
- Setup Wizard
- Product List
- Product Rules

If the active product does not require setup:

- Changeover
- Auto Setup
- Product List
- Product Settings
- Product Rules

5.1.2.1 Changeover Page

Metal Detector	Access Level Required
All Metal Detectors	Operator, Supervisor, Engineer or Administrator

Choose the active product.

The Changeover page shows all products in the system and allows you to select a new active product.

Each product is shown with its Name, product code and Active status.

You can filter the list using the funnel icon .

Note The current active product is grayed out - you cannot select it since it cannot replace itself.

Note You can access the Changeover page from everywhere in the UI by tapping the Active Product area in the top right corner of the screen.

See also

 Viewing and Changing Products ▶ Page 64

5.1.2.2 Auto Setup Page

Metal Detector	Access Level Required
All Metal Detectors	Engineer, Supervisor or Administrator

Run a routine to automatically set inspection settings.

Note This page remains hidden in the Products section until the active product has completed the initial setup process.

On the Auto Setup page you can:

- use the Mode Selection drop-down to set an operating frequency for the metal detector.
- start an automatic product setup process, which calibrates the machine to the product to be scanned.
- tap the cog icon  to access the Auto Setup Settings section, where you can fine-tune the setup.

See also

 Auto Setup Settings Pages ▶ Page 27

 Setting up Products ▶ Page 62

 Understanding Choice of Operating Mode ▶ Page 14

5.1.2.3 Setup Wizard Pages

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

The active product requires setup. Set up the inspection and rejection settings.

The Setup Wizard walks you through the process of initial product set up.

Note This page is shown only when the active product requires initial set up.

See also

 Understanding Phase and Amplitude ▶ Page 13

 Understanding Choice of Operating Mode ▶ Page 14

 Setting up Products ▶ Page 62

5.1.2.4 Product List Page

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator (Supervisor has some limited access to product settings)

Create, edit, copy, and delete products.

Each product is shown with its Name, product code and Active status.

You can filter the list using the funnel icon .

Select New to create an item from scratch.

Select an existing item to open a menu, allowing you to:

- Carry out a product Changeover (unless you select the currently active product).
- Use the Setup Wizard to set up a product.
- Edit the item's Settings.
- Copy the item to create a new item.

- Delete the item.

See also

- [Changeover Page ▶ Page 18](#)
- [Viewing and Changing Products ▶ Page 64](#)
- [Product Settings Pages ▶ Page 20](#)
- [Setting up Products ▶ Page 62](#)

5.1.2.5 Product Settings Pages

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator (Supervisor has some limited access to product settings)

Edit the active product inspection and rejection settings.

Note This page remains hidden in the Products section until the active product has completed the initial setup process.

This section contains the following pages:

- Identification - edit the product Name and Code.
- Inspection Settings - change Mode, Sensitivity and Phase. From this page you have shortcuts to:
 - Auto Setup
 - Advanced Settings - a page where you can access further setup options: Mode, Sensitivity 1, Sensitivity 2, Phase and Phase Tolerance.
- Reject Settings - set up and enable Synchronized Reject.
Set up reject behavior: Detection to Rejection interval and Reject On Extension.
- Advanced - set up Pack Sensor Blocked Time interval.

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Engineer or Administrator
StandardLine - M31R with PlusLine Productivity Pack	(Supervisor has some limited access to product settings)

PlusLine and StandardLine systems with the Productivity Pack installed, have the following additional feature:

- Test Samples
 - Enable the tick box to use the system Test Sample settings.
 - Disable the tick box to define test sample sizes for the individual product in the same way as described on the QA > Test Sample Setup page.

See also

- [Understanding Phase and Amplitude ▶ Page 13](#)
- [Phase Control ▶ Page 13](#)
- [Understanding Choice of Operating Mode ▶ Page 14](#)
- [Auto Setup Page ▶ Page 19](#)
- [Setting up Products ▶ Page 62](#)
- [Test Sample Setup ▶ Page 70](#)

5.1.2.6 Product Rules Page

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

Set rules for product identification.

The Product Rules page allows you to determine whether product codes are mandatory.

- This is a mandatory setting for ProdX, and will be auto-enabled if required.

- 1 Navigate to **Products > > Product Rules**.
- 2 Select or clear the **Mandatory Product Code Enabled** option as required (default clear).

Note Enabling this setting will affect existing product identification.

See also

- [Managing Products ▶ Page 62](#)
- [ProdX Settings Page ▶ Page 29](#)

5.1.3 Statistics Pages

The Statistics pages allow you to view production statistics and modify statistics-related settings.

5.1.3.1 Production Page

Metal Detector	Access Level Required
All Metal Detectors	All Users

View and reset production counters.

The Production page shows production counters, Rejected and Total.

If you are logged in as Administrator, Engineer, QA or Supervisor, you can also reset the counters.

5.1.3.2 Settings Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

View and modify statistics-related settings.

This section contains the following pages:

- Counters - allows you to control whether statistics counters will automatically reset when the active product is changed.
Select or clear the Reset Counters on Product Changeover option respectively.
- Detection Log - allows you to group by product and clear detection logs.

See also

- [Changeover Page ▶ Page 18](#)
- [Viewing and Resetting the Counters and Settings ▶ Page 68](#)

5.1.3.3 Detection Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	All Users
StandardLine - M31R with PlusLine Protection Pack	

View historical detection.

This page allows you to view the detection history for all products.

You can filter the list using the funnel icon .

Select the cog icon  to access the Detection Log page to group or clear detection logs.

See also

- [Settings Page ▶ Page 21](#)

5.1.4 QA Pages

The QA pages allow you to set up Performance Tests, manage and download the Performance Test History, and view the Access Log.

5.1.4.1 Performance Test Page

Reset Performance Test due time.

The Performance Test page allows you to log a Reminder Only Test and/or Standard Test, where appropriate.

The feature remains inactive until the required performance test has been enabled in the QA > Test Setup page.

Two types of test are available:

- Reminder Only Test
- Standard Test

See also

[Test Setup Pages](#) ► Page 22

5.1.4.1.1 Reminder Only Test

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

Note If required, individual users can be granted access rights to test options.

This page assumes a test has already been completed. Select Pass to log a successful test.

If configured to run tests at regular intervals, this will reset the test interval: the Next Test Due date and time will change as configured in Test Setup.

A conveyor override option is available allowing the conveyor to continue operating in the event a scheduled test being overdue.

Note If tests are configured to run at regular intervals, when a test becomes due, Next Test Due changes to Test Due and the background changes to amber.

See also

[Test Setup Pages](#) ► Page 22

5.1.4.1.2 Standard Test

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

Note If required, individual users can be granted access rights to test options.

The Standard Test enables you to test separately for the following types of contaminant:

- Ferrous - Red
- Non-Ferrous - Yellow
- Stainless Steel - Blue

The page for each individual test contains a Pass, Fail and Cancel button.

If extra due diligence sensors are configured (Rejected Product Sensor and Reject Check Sensor), then these must also operate correctly for a Pass to be registered.

You cannot log a successful test until the system has detected and rejected each active contaminant (see Test Sample Setup). All Due/Overdue tests need to pass, the alarm is then cleared automatically.

Note The material color code is shown during the Standard Test performance test.

See also

[Test Setup Pages](#) ► Page 22

[Test Sample Setup Page](#) ► Page 23

5.1.4.2 Test Setup Pages

Set the performance test schedule.

Two types of test are available:

- Reminder Only Test
- Standard Test

Reminder Only Test

Metal Detector	Access Level Required
StandardLine - M31R	Administrator

Options available:

- Enabled
- Scheduling Mode
- Due Interval
- Schedule Type
- Login Required

Reminder Only Test and Standard Test

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Productivity Pack	

Options available:

Scheduled Settings -

- Enabled
- Scheduling Mode
- Due Interval
- Schedule Type
- Overdue Now
- Stop When Overdue

Triggers -

- Test Following Product Change
- Test Following Settings Change
- Test Following Start-Up

Access Control -

- Login Required

See also

[Performance Test Page](#) ► Page 21

[Managing Performance Test Access](#) ► Page 37

5.1.4.3 Test Sample Setup Page

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Protection Pack	

This page allows you to enable and configure the test sample size for the following materials:

- Ferrous - **Red**
- Non-Ferrous - **Yellow**
- Stainless Steel - **Blue**

Current product Test Sample settings will override settings configured on this page. If configured within the product settings, a message will be displayed on the UI to link the user back to the Product Setting page.

Note The material color code is shown during the Standard Test performance test.

5.1.4.4 History Page

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

View performance test history.

This page contains the list of all tests logged into the system.

The tests are grouped by product and by date, and are listed chronologically.

You can filter the list using the funnel icon .

You can access the history download page using the download icon .

For each test, you can see:

- The time when the test was logged.
- The type of test - the available options will depend on the metal detector model.
 - For Standard Tests only, select the blue arrow to display samples tested, their results, and signal strength.
- The result - passed or failed.
- The user who logged the test.

Note For StandardLine systems, when a test is downloaded, it is removed from the log.

5.1.4.5 Reports

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

The Reports page allows you to download the following test data:

- Performance Test - Generate and download performance test reports.
- User Access - Generate and download access log reports.
- Alarms - Generate and download alarm log reports.
- Detections - Generate and download detection log reports.

Select the desired option to display the log report on the UI screen.

The reports are grouped by date and arranged chronologically.

You can filter the list using the funnel icon .

Expand and collapse the log results using the  icon.

Instead of selecting individual items or groups of items on the list, you can select all items by using the checkbox icon  at the top of the page:

- Tap the icon once to select all new reports.
- Tap the icon a second time to include all new and downloaded reports.
- Tap the icon a third time to deselect all reports.

Note The checkbox icon displays changes depending on the current selection or options available.

See also

 Event Logs and Reports ► Page 74

5.1.4.6 Settings

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Protection Pack	

View and modify the settings related to the performance test history.

Use the tick box to enable and disable the Performance Test History Full alarm function.

See also

 Performance Test Page ► Page 21

 Reports ► Page 24

 Downloading the Performance Test ► Page 80

5.1.4.7 Access Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

View historical user activity.

Date and time are noted against user activity - login and out.

You can filter the list using the funnel icon .

Select the cog icon  to access Access Log Settings where Administrators can clear the access logs.

See also

 [Access Log](#) ▶ Page 27

 [Access Log](#) ▶ Page 77

5.1.5 System Pages

The System section of the UI provides access to various system settings and system information.

There are multiple pages in this section. Each group of settings can be accessed from the main System pages:

- Information, Region Settings, Users, and Auto Setup Settings.
- Speed Settings, System Integration, Detection and Rejection, and Communications.
- ProdX Settings, Licensed Options, Display and Buzzer and Alarm Settings.
- Alarm Log, Diagnostics, Self Service, and Identification.

Note Not all features will be present on all systems, or located as indicated above.

5.1.5.1 Information Pages

Metal Detector	Access Level Required
All Metal Detectors	All Users

View serial number, software version, date/time, and other key information

This section contains the following pages:

- Basic, offers an overview of the installed system:
 - Serial Number
 - Head Type
 - Firmware Version
 - Mode
 - Model Type
 - Date/Time
- Licensing, shows the following information:
 - Serial Number
 - License Key
 - Host ID
- OS, provides information about the installed operating system:
 - OS Version
 - OS Update Version
 - Base Software Image Version
- Network Info, provides information about the network connection:
 - IP Address
 - Subnet Mask
 - Default Gateway
 - DHCP
 - Network Cable

5.1.5.2 Region Settings Pages

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Language and date/time settings.

This section contains the following pages:

- The Preferences page allows you to change the displayed system language and units of measure.

Options available:

- Language - change the system language for all users.
- Measurement Units - select metric or imperial units of measure.

Note Logged in users can change their UI language from the My Account page.

- The Date and Time page allows you to set the system date and time according to your local time zone.

Options available:

- Time Zone
- Date
- Date Format
- Time
- 24-hour display

Note If you change the time zone, you will be prompted to restart the system.

See also

- [Setting the System UI Language ▶ Page 38](#)
- [Setting the System Date and Time ▶ Page 38](#)
- [My Account Pages ▶ Page 26](#)

5.1.5.3 Users Pages

Manage user accounts and access rights

This section contains the following pages:

- Accounts
- My Account
- Access Log
- Access Log Settings

5.1.5.3.1 Accounts Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Create, edit, and delete user accounts.

Each user is shown with name, role and active status.

You can filter the list using the funnel icon .

Select New to create an item from scratch.

Select an existing item to open a menu, allowing you to:

- Edit the item's Settings.
- Copy the item to create a new item.
- Delete the item.

See also

- [Administrators: Managing User Accounts ▶ Page 35](#)
- [Managing Performance Test Access ▶ Page 37](#)

5.1.5.3.2 My Account Pages

Metal Detector	Access Level Required
All Metal Detectors	All logged in users

Edit settings for logged in user account.

This section contains the following pages:

- Details - change your password and UI language.

You can also view your account Role and Enabled status but you cannot change these settings.

Note In addition to the above, if you are an Administrator, you can also edit your account Name from this page.

- Additional Privileges - select or clear the Allow Performance Test Access option.

Note If you are an Administrator, this option cannot be cleared.

See also

[Changing your Account Settings](#) ▶ Page 34

5.1.5.3.3 Access Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

View historical user activity.

Date and time are noted against user activity - login and out.

You can filter the list using the funnel icon .

Select the cog icon  to access Access Log Settings where Administrators can clear the access logs.

See also

[Access Log Settings](#) ▶ Page 27

[Access Log](#) ▶ Page 77

5.1.5.3.4 Access Log Settings

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Protection Pack	

View and modify Access Log related settings.

This page allows the administrator to clear access logs.

5.1.5.4 Auto Setup Settings Pages

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set setup level, product type, and if reject should be inhibited.

This section contains the following pages:

- Auto Setup Settings, contains the following settings:
 - Signal Level for Auto Setup
 - Product Type
 - Material Optimization
 - Reject Inhibit
- Vibration Limited Sensitivity, contains the following settings:
 - Enable Vibration Limited Sensitivity
 - Vibration Limited Sens - a list of frequency sensitivity values relevant to your system.

5.1.5.5 Speed Settings Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set fixed or variable speed, and expected product speed.

This page allows you to change the Speed Setting: choose either Variable or Fixed.

If you choose Variable, you can also set Maximum Speed and you are also shown the Operational Speed Range system value.

If you choose Fixed, you can set System Speed or start the Auto Speed feature to calculate the value automatically.

See also

[Speed Settings](#) ► Page 54

5.1.5.6 System Integration Pages

Metal Detector	Access Level Required
All Metal Detectors	Administrator

This section contains the following pages:

- Inputs - see and configure available inputs.
- Outputs - see and configure available outputs.
- Conveyor
- PackML - enable Show PackML State.

5.1.5.6.1 Inputs Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Assign functions to custom inputs.

The page shows the list of all inputs available in the system.

Each item is shown with Function, Location, and Active status.

You can filter the list using the funnel icon .

Select an item to access the following settings:

- Function
- Active Level

See also

[Managing Inputs](#) ► Page 41

5.1.5.6.2 Outputs Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Assign functions to custom outputs.

The page shows the list of all outputs available in the system.

Each item is shown with Function, Location, and Active status.

You can filter the list using the funnel icon .

Select an item to access the following settings:

- Function
- Active Level

See also

[Managing Outputs](#) ► Page 48

5.1.5.6.3 Conveyor Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set how the conveyor is controlled.

This page provides access to the following settings:

- Start/Stop Control
- Start Guard Enabled

See also

[Conveyor Start/Stop Control Options](#) ► Page 55

5.1.5.6.4 PackML Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set if PackML state should be visible.

This page provides access to the following settings:

- Show PackML State
 - When the box is checked, a machine status message is shown below the small bargraph in the right-hand pane of most screens.

Note If Bin Unlock is configured, the PackML state will be hidden by the "Bin Unlock" button.

See also

[Machine Status \(PackML State\)](#) ► Page 56

5.1.5.7 Detection and Rejection Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Configure Detection and Rejection behavior.

This page provides access to the following settings:

- Reject Inhibit
- Unsynchronized Detection Events Enabled
- Changeover Rejection Enabled
- Reject Confirmation Enabled

5.1.5.8 Communications Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set IP address and other network settings.

This page provides access to the following settings:

- DHCP
- IP Address
- Default Gateway
- Subnet Mask

Note If DHCP is switched on, the rest of the values are assigned automatically and will appear read-only.

5.1.5.9 ProdX Settings Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Setup ProdX Communication.

This page provides access to the following settings:

- ProdX Enabled
- Device Server Port
- ProdX Server IP
- ProdX Server Port
- Disconnect Action

See also

[ProdX](#) ► Page 59

5.1.5.10 OPC UA

Metal Detector	Access Level Required
All Metal Detectors with OPC UA License	Administrator

OPC UA is a communications protocol that grants access to information contained within the metal detector permitting a secure and reliable exchange of data between the metal detector and external device.

An additional license is required for the activation of the OPC UA server. This license can be purchased at time of order with the metal detector, or separately at a later date.

For information on Setup and Operation of the OPC UA Server, please refer to the M-Series OPC UA Server Technical Manual for Engineering & IT: <http://TO BE CONFIRMED>

For further information about this upgrade, please contact your METTLER TOLEDO sales representative.

5.1.5.11 Licensed Options Pages

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set license key and view licensed options.

This section contains the following pages:

- License Key
View the Serial Number and Host ID of your product, and enter the License Key.
- Licensed Options
View the licenses available to your system, including the line type and compatible feature packs.

See also

 Technical Specifications ► Page 92

5.1.5.12 Display and Buzzer Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Calibrate touchscreen, set brightness, and disable buzzer.

This page provides access to the following settings:

- Calibrate Touchscreen
- Brightness
- Buzzer Enabled

Note Refer to the M30R-Series Installation Manual for further information on calibrating the touchscreen.

5.1.5.13 Alarm Settings Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set alarm display priority and auto acknowledgment.

This section contains the following pages:

- Alarm Settings, Basic, contains the following settings:
 - Display Priority
 - Local Acknowledge Access
 - Local Reset Access
 - Acknowledge Inactive Alarms Automatically
- Alarm Settings, Alarm Log, contains the following settings:
 - Clear Alarm Log.

See also

 Alarms Page ► Page 31

5.1.5.14 Alarm Log

Metal Detector	Access Level Required
All Metal Detectors	All Users

View historical alarms.

This page allows you to view the historical Alarm Log.

You can filter the list using the funnel icon .

Select the cog icon  to access the Alarm Log page to clear alarm logs, see System > > Alarm Settings > Alarm Log.

See also

 Alarm Settings Page ► Page 30

5.1.5.15 Diagnostics Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Copy diagnostics logs to USB stick.

This page provides access to the following settings:

- Logging Level
- Copy Logs to USB

5.1.5.16 Self Service Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Service the metal detector.

This page provides a Serial Number and Challenge Code to pass on to a METTLER TOLEDO Service representative in case you need to access self service functionality, for example to update the firmware.

See also

 System Updates ► Page 82

5.1.5.17 Identification Page

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Set the Line ID.

This page contains one setting, Line ID.

5.1.6 Alarms Page

Metal Detector	Access Level Required
All Metal Detectors	All Users

The Alarms page contains information about all current alarms in the system and enables you to acknowledge alarms.

If no alarms are Active, the page shows a System Healthy message.

If one or more alarms are Active, the page shows the list of current alarms.

Each alarm is shown with its Name, Code, Timestamp, Active status and severity icon. You can acknowledge alarms straight from this list. You can also use the checkbox icon  at the top of the page to acknowledge all alarms at once.

Select an individual alarm to open and review it.

Each alarm contains the following pages:

- Summary

- Details
- Timestamp
 - Contains alarm log in case of supported licenses.

Depending on the logged in user, you may see a cog icon  at the top of the page.
Select the cog icon  to access the Alarm Settings page.

See also

 [Alarm Settings Page](#) ▶ Page 30

5.2 Keyboards and Character Sets

The on-screen virtual keyboard may change its layout depending on the language used.
Currently the following language layouts are supported:

- English (default)
- German
- French
- Spanish
- Czech
- Polish
- Hungarian

Press and hold any key on the keyboard to view and select accented characters in your language, such as:

- á, è, ü, š, etc.

The language can be set up for the entire system or just for the logged in account.

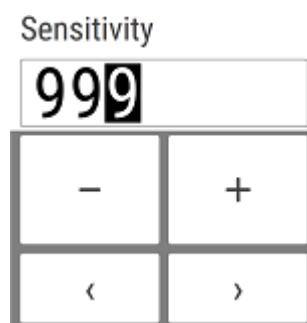
See also

-  [Changing your Account Settings](#) ▶ Page 34
-  [Changing the UI Language for a Specific Account](#) ▶ Page 36
-  [Setting the System UI Language](#) ▶ Page 38

5.3 Nudge Editors

A nudge editor has four keys that allow you to increase or decrease a numerical value in large or small steps.
This is useful for live adjustments, for example when you are manually setting the Sensitivity for an active product.

Either select the digit or use the arrow keys   to highlight the digit you want to change, and then use the **+** and **-** keys to nudge the value up or down.



6 All Users: Managing your Account

6.1 User Accounts and Roles

User accounts belong to one of several roles, which determine the user's access level:

- An Administrator has access to everything in the system, including the management of users, the metal detector's interaction with the conveyor system, and the metal detector's interaction with external devices.
- An Engineer manages product configuration.
- An Operator performs day-to-day tasks.
- A Supervisor has access to product set-up and settings.
- A QA is allowed to run Performance Test tasks.
- A guest user (not logged in) can only access some information about the product.

When you are logged in with a role other than Administrator, you will see menus with a padlock icon:



The padlock icon means your access is restricted.

6.2 Default Accounts and Passwords

The product is delivered to your site with default accounts, one for each role. This enables you to log in as any role using the default passwords:

- For Administrator, the password is 9999.
- For QA, the password is 5555.
- For Engineer, the password is 3333.
- For Supervisor, the password is 2222.
- For Operator, the password is 1111.

Note Administrators can Create, Edit and Delete any account (apart from their own) and change any password. Other users are restricted to changes within their own accounts.

See also

- [User Accounts and Roles ▶ Page 33](#)
- [Changing your Account Settings ▶ Page 34](#)
- [Administrators: Managing User Accounts ▶ Page 35](#)
- [Logging In ▶ Page 33](#)

6.3 Logging In

Metal Detector	Access Level Required
All Metal Detectors	All Users

Note Individual user accounts are set up by an Administrator.

To log in:

- 1 Select **Log In**.
- 2 On the **Select account** page, select the account you wish to log into.
- 3 On the **Enter Password** page, type your password.
- 4 **Accept**.

See also

- [Administrators: Managing User Accounts ▶ Page 35](#)

6.3.1 Automatic Logout

You will be logged out automatically following a 5-minute period of inactivity. The green bar under your **Username** decreases from right to left, to give an indication of time remaining.

6.4 Changing your Account Settings

Metal Detector	Access Level Required
All Metal Detectors	All Users

As a logged in user, you can change your own password and language. These settings are available on the My Account > Details page.

You can also see whether you have access to performance tests - this setting is available on the My Account > Additional Privileges page.

Note Only an Administrator can change this setting, and only for QA, Supervisor, Engineer and Operator accounts.

- 1 Navigate to **System > Users > My Account**.
- 2 **Edit** as required.
- 3 **Accept**.

Note In addition to the above, if you are an Administrator, you can also edit your account Name from this page.

7 Administrators: Managing User Accounts

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Administrators can Create, Edit and Delete any account (apart from their own) and change any password. Other users are restricted to changes within their own accounts.

Administrators can also allow individual accounts to access performance tests, overriding the default access level.

You can have multiple users per role but you cannot add or remove roles.

7.1 Creating an Account

Metal Detector	Access Level Required
All Metal Detectors	Administrator

On the System > Users > Accounts page you can create a new account from scratch, or copy an existing account.

To create an account from scratch:

- 1 Select **New**.
 - ➔ The **Account Settings > Details** page opens.
- 2 Enter **Name**, **Password**, **Language** and **Role** for the new account. If required, you can also disable the new account by deselecting the **Enabled** check box.
- 3 If required, navigate to the next page, **Additional Privileges**, and configure **Allow Performance Test Access** accordingly.

Note You cannot edit Allow Performance Test Access if you are creating an Administrator account.

To copy an existing account:

- 1 Select the account you wish to copy, select **Copy**.
 - ➔ A new account is created, a replica of the original. It has the same name as the original account, followed by "- Copy".
- 2 Select the new account and then select **Settings**.
 - ➔ The account opens for editing.
- 3 Enter **Name**, **Password**, **Language** and **Role** for the new account. If required, you can also disable the new account by deselecting the **Enabled** check box.

7.2 Renaming an Account

Metal Detector	Access Level Required
All Metal Detectors	Administrator

An Administrator can rename any account, including the logged in account.

To change:

- 1 Navigate to **System > Users > Accounts**.
- 2 Select the account you wish to change and choose **Settings**.
- 3 Select the **Name** field.
 - ➔ The **Name** page opens.
- 4 **Edit** as required.
- 5 **Accept**.

Note If you are logged in as Administrator you can also rename your own account from the My Account page.

See also

 Changing your Account Settings ► Page 34

7.3 Changing Passwords

Metal Detector	Access Level Required
All Metal Detectors	Administrator

To change:

- 1 Navigate to **System > Users > Accounts**.
- 2 Select the account you wish to change and choose **Settings**.
- 3 Select the **Password** field.
- 4 **Edit** as required.
- 5 **Accept**.

Note All logged in users, regardless of their role, can change their own account password from the My Account page.

See also

- [Changing your Account Settings ▶ Page 34](#)
- [Default Accounts and Passwords ▶ Page 33](#)

7.4 Changing the UI Language for a Specific Account

Metal Detector	Access Level Required
All Metal Detectors	Administrator

An Administrator can determine the UI language for any user account.

To change:

- 1 Navigate to **System > Users > Accounts**.
- 2 Select the account you wish to change and choose **Settings**.
- 3 Open the **Language** drop-down and select a new language from the list.

Note Logged in users can change their UI language from the My Account page.

Note An Administrator also sets up the default UI language for the system. This functionality is available from System > Region Settings. Accounts will take the default setting unless their language has been set up explicitly.

See also

- [Changing your Account Settings ▶ Page 34](#)
- [Setting the System UI Language ▶ Page 38](#)

7.5 Changing the Account Role

Metal Detector	Access Level Required
All Metal Detectors	Administrator

An Administrator can change the account role, selecting from a predefined list of available roles.

You can have multiple users per role but you cannot add or remove roles.

Note It is not possible to change the role of the currently logged in account.

To change:

- 1 Navigate to **System > Users > Accounts**.
- 2 Select the account you wish to change and choose **Settings**.
- 3 Tap the **Role** drop-down and select a new role from the list.

See also

- [User Accounts and Roles ▶ Page 33](#)

7.6 Enabling and Disabling an Account

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Accounts can be enabled and disabled as required. A disabled account disappears from the list of accounts available for login (Log In > Select account) but remains in the list of accounts available to an Administrator for editing (System > Users > Accounts).

To change:

- 1 Navigate to **System > Users > Accounts**.
- 2 Select the account you wish to change and choose **Settings**.
- 3 Select or clear the **Enabled** check box as required.

7.7 Managing Performance Test Access

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Access to performance tests is set in two ways:

- On an Administrator account, this option is always selected and cannot be cleared.
- Within an individual user account, an Administrator can configure access for that particular account. Access to performance tests will be granted even if the account is not Administrator or QA.

To configure performance test access for an individual account:

- On the QA > Test Setup >> Access Control page, if Login Required is selected, only Administrator and QA users will be able to access performance tests. Otherwise anyone can access performance tests, including not logged in users.
- Open the account for editing and on the second page, Additional Privileges, select or clear Allow Performance Test Access accordingly.

See also

[Accounts Page](#) ► Page 26

[Test Setup Pages](#) ► Page 22

7.8 Deleting an Account

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Note You cannot delete the account you are logged into.

To delete:

- 1 Navigate to **System > Users > Accounts**.
- 2 Select the account you wish to delete and choose **Delete**.

8 Setting System Language, Date and Time

8.1 Setting the System UI Language

Metal Detector	Access Level Required
All Metal Detectors	Administrator

When the required system language has been selected, all UI menus and screens will be displayed in that language, except where the UI language has been individually defined for a specific user account.

To change:

- 1 Navigate to **System > Region Settings**.
- 2 Select required **Language** from the drop-down list.
- 3 **Accept**.

Both an Administrator and the logged in user can change the UI language for a specific account.

See also

- [Changing the UI Language for a Specific Account ▶ Page 36](#)
- [Changing your Account Settings ▶ Page 34](#)

8.2 Setting the System Date and Time

Metal Detector	Access Level Required
All Metal Detectors	Administrator

To change:

- 1 Navigate to **System > Region Settings >> Date and Time**.
- 2 Set the following as required:
 - ➔ **Time Zone**
 - ➔ **Date**
 - ➔ **Date Format**
 - ➔ **Time**
 - ➔ **24-hour display** - select and clear the check box to toggle between a 12-hour and a 24-hour display

Note If you change the time zone, you will be prompted to restart the system.

9 System Integration

9.1 Programmable Inputs and Outputs

The M30 R-Series Installation Manual provides details of the connections to the CPU I/O board and technical specification requirements.

M30 R-Series metal detectors have ten programmable inputs and six programmable outputs as standard.

These inputs and outputs allow the metal detector to interact with conveyor systems, devices, controllers, and sensors.

The inputs and outputs often work in pairs. For example, one of the inputs and one of the outputs are connected to the door of the reject bin, which means that the input informs the metal detector that the reject bin is open. An output from metal detector can lock and unlock the reject bin.

Each tile in the Inputs and Outputs lists in System Integration indicates Active or Inactive followed by High or Low. The High or Low refers to the electrical signal state of the Input or Output. Active High and Active Low are physical states which interact with two logical states Active and Inactive.

- When an input is activated, a message on the screen is sometimes generated (if applicable) from the selected option (for example "Air Supply Fail").
- Inputs influence the logic and behavior of the metal detector in different ways.

Note The option to add an additional six programmable inputs and six programmable outputs is available with the integration of the IO Expansion Board (optional).

Note It is advised any safety, speed, or reject critical devices are not connected through the IO Expansion Board.

Note It is not possible to connect the Speed Control Sensor to the Inputs of the IO Expansion Board.

See also

- [Managing Inputs ▶ Page 41](#)
- [Managing Outputs ▶ Page 48](#)

9.1.1 Programmable Input List

The following is a list of typical functions that can be assigned:

- [Unused ▶ Page 42]
- [Acknowledge All Alarms Input ▶ Page 42]
- [Acknowledge And Reset All Alarms Input ▶ Page 42]
- [Air Supply Sensor Input ▶ Page 43]
- [Bin Full Sensor Input ▶ Page 43]
- [Bin Unlocked Input ▶ Page 44]
- [Conveyor Running Input ▶ Page 44]
- [Conveyor Start Input ▶ Page 44]
- [Conveyor Stop Input ▶ Page 44]
- [Downstream Blocked Input ▶ Page 45]
- [Pack Sensor Input ▶ Page 45]
- [Reject Check Sensor Input ▶ Page 45]
- [Reject Operation Sensor Input ▶ Page 45]
- [Rejected Product Sensor Input ▶ Page 46]
- [Reset All Alarms Input ▶ Page 46]
- [Safety Circuit Input ▶ Page 47]
- [Speed Sensor Input ▶ Page 47]
- [Upstream Starved Input ▶ Page 47]

9.1.2 Programmable Output List

The following is a list of typical functions that can be assigned:

- [Unused ▶ Page 48]

- [Air Supply Failure Output ▶ Page 48]
- [Bin Full Output ▶ Page 48]
- [Bin Unlock Output ▶ Page 49]
- [Bin Unlocked Output ▶ Page 49]
- [Conveyor Drive Output ▶ Page 49]
- [Current Alarms Output ▶ Page 49]
- [Current Fault Alarms and Higher Output ▶ Page 50]
- [Current Warning Alarms Output ▶ Page 50]
- [Current Warning Alarms and Higher Output ▶ Page 50]
- [Detect Output ▶ Page 50]
- [Good Pack/Product Output ▶ Page 51]
- [Latched Alarms Output ▶ Page 51]
- [Performance Test Due Output ▶ Page 51]
- [Performance Test Overdue Output ▶ Page 51]
- [ProdX Disconnected Output ▶ Page 51]
- [Reject Output ▶ Page 51]
- [Reject Not Confirmed Output ▶ Page 52]
- [Stop On Detection Output ▶ Page 52]
- [System Healthy Output ▶ Page 52]
- [System Stop Output ▶ Page 53]
- [Unacknowledged Alarms Output ▶ Page 54]

9.1.3 Input and Output Configuration

9.1.3.1 Input Configuration

Each programmable input has two activation levels:

- Active High
- Active Low

The activation level can be selected from the UI once the system has booted up.

Active High

Configure the programmable input to Active High when the following is true:

- The input is normally switched ON when the system is powered ON.
- The default signal from the device connected to the input terminal is high (+24 V) in PNP configuration or low (0 V) in NPN configuration.
- The normal state of the programmable input status LED is ON.
- When the input device is activated/triggered, the signal to the input terminal drops out and the programmable input status LED is OFF.
- Example input devices with Active High configuration:
 - Volt free Normally Closed contact (for example Stop push button or E-Stop).
 - Photo electric sensor (Light On type).

Active Low

Configure the programmable input to Active Low when the following is true:

- The input is normally switched OFF when the system is powered ON.
- The default signal from the device connected to the Input terminal is low (0 V) in PNP configuration or high (+24 V) in NPN configuration.
- The normal state of the programmable input status LED is OFF.
- When the device is activated/triggered, the signal to the input terminal goes high and the programmable input status LED is ON.
- Example input devices with Active Low configuration:
 - Volt free Normally Open contact (for example Start or System Reset push button).
 - Photo electric sensor (Dark On type).

9.1.3.2 Output Configuration

Programmable Outputs 1 to 4 have two activation levels:

- Closed
- Open

The activation level can be selected from the UI once the system has booted up.

Active Level - Closed

Select the programmable output Active Level to Closed when the output device needs to be activated when the condition is true.

For example, operating an audible alarm when there is a detection.

Active Level - Open

Select the Programmable output Active Level to Open when the output device needs to be de-activated when the condition is true.

For example, when the fault alarm (normally closed) signal from detector is used to run the downstream and upstream systems.

Programmable Output 5 and 6 have two activation levels:

- De-energized
- Energized

The output activation levels are not configurable. The default values can be obtained from the UI when an output function is selected.

Active Level – De-Energized

The output relay de-activates when the output function is true.

For example, the Reject Bin Full signal from the normally closed contact for programmable output relays 5 or 6.

Note When the Bin Full signal is Active the output relay de-energizes hence the reject Bin Full beacon will be ON if connected to the normally closed contact.

Active Level – Energized

The output relay activates when the output function is true.

For example, the drive enable signal from the normally open contact for programmable output relays 5 or 6.

Note When the system is healthy, and the drive enable signal is Active, the output relay energizes. The motor run signal will be Active if connected to the normally open contact.

9.2 Managing Inputs

The availability of the following features will depend on the individual metal detector specification. Depending on the features selected, other associated features may be hidden.

9.2.1 Setting up Inputs

Metal Detector	Access Level Required
All Metal Detectors	Administrator

Note METTLER TOLEDO offers pre-configured settings for commonly used systems. The configuration you need may already have the correct system integration in place.

The Inputs page shows all available inputs as tiles. Each tile contains the following information:

- The **Title** is the assigned function (default - Unused).
 - Select the tile that the external device you want to set up is connected to on the CPU I/O board.
- The **I/O board** where the function is located:
 - Main I/O = CPU I/O board.
 - Serial No = Expansion I/O board (optional).
- The **Active status** of the input.
 - Refer to Input/Output configuration for further details.

1 Navigate to **System > > System Integration > Inputs**.

- 2 Select required Tile.
- 3 Select the required **Function** from drop-down.
- 4 **Edit** the parameters as required.
 - ➔ The number of cells will vary depending on the device being set up.
- 5 Set **Active Level** as required.

See also

- [Input Configuration](#) ▶ Page 40
- [Managing Inputs](#) ▶ Page 41

9.2.2 Unused

These are spare ports.

The location of all the ports is indicated:

- Main I/O (CPU I/O board) 1 to 10.
- Where fitted "I/O expansion board serial Number" - 1 to 6.

9.2.3 Acknowledge All Alarms Input

When activated, this input function will acknowledge any alarms that require acknowledgment.

Alarms requiring acknowledgment are used to ensure that a condition has been communicated to the user, typically a condition of particular importance. The alarm can still remain active, even after acknowledgment.

Latchable alarms remain active even when the underlying cause has cleared. Such alarms will remain latched indefinitely and must be reset explicitly. A latchable alarm can only be reset when the alarm input condition is removed. Once a latchable alarm is reset, it is unlatched and becomes inactive.

Alarms requiring acknowledgment allow the system to behave differently once the alarm has been acknowledged by the user, even though the underlying condition is still active

- For example, a siren might sound when the alarm is unacknowledged and then deactivate when the alarm is acknowledged.

Each individual alarm will have its own default configuration (either acknowledgment required or acknowledgment not required).

By separating acknowledge and reset functionality, it is possible, for example, to provide open access to acknowledge functionality (for example via a push button) whilst limiting access to the reset function (for example via a reset key).

Options

There are no options associated with this function.

See also

- [Reset All Alarms Input](#) ▶ Page 46
- [Managing Alarms](#) ▶ Page 84

9.2.4 Acknowledge And Reset All Alarms Input

Separating the acknowledge and reset functionality allows you, for example:

- To provide open access to acknowledge functionality (via a push button), while limiting access to the reset function (via a reset key).

On simpler systems, it is often more convenient to combine the acknowledge and reset functions into a single push button or key.

Latchable alarms remain active even when the underlying cause has cleared. Such alarms will remain latched indefinitely and must be reset explicitly. A latchable alarm can only be reset when the alarm input condition is removed. Once a latchable alarm is reset, it is unlatched and becomes inactive.

Options

There are no options associated with this function.

See also

- [Reset All Alarms Input](#) ▶ Page 46
- [Managing Alarms](#) ▶ Page 84

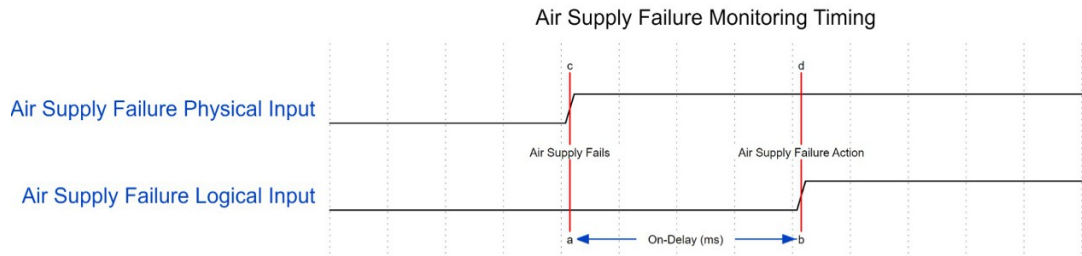
9.2.5 Air Supply Sensor Input

The Air Supply Sensor input function can be used to warn operators and stop the system when air pressure is low.

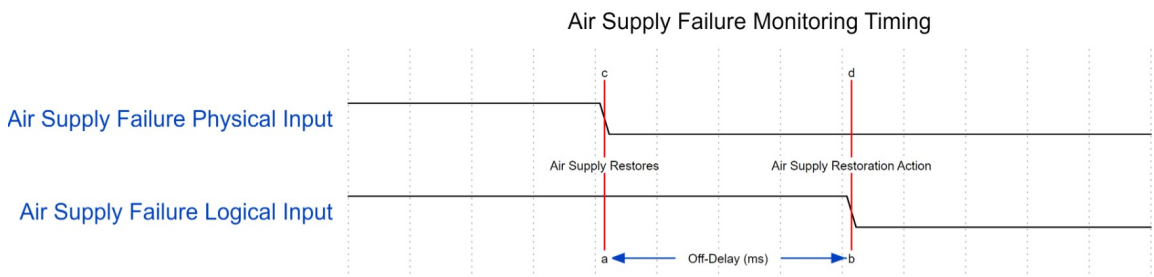
Options

There are three options associated with this function:

- On Delay - is a configurable period of time which will delay activation of the input function and suppress any spurious input signals.



- Off Delay - is a configurable period of time which will delay activation of the output function and suppress any spurious output signals.



- Action - will configure the behavior of the system when the input function activates:
 - Immediate Stop - activating the input will immediately activate the fault alarm and stop the system.
 - Stop on Next Reject - activating the input will initially activate the warning alarm, but will not stop the system. When the next reject is scheduled, the fault alarm will activate and stop the system.

See also

[Air Supply Failure Output](#) ► Page 48

9.2.6 Bin Full Sensor Input

The Bin Full Sensor function should be used to generate an alarm when the product reject bin becomes full. When the Bin Full output is configured, it will activate.

Options

There are three options associated with this function:

- On Delay is a configurable period of time which the sensor needs to be active before the Bin Full condition is satisfied. This allows rejects to pass the sensor without activating the alarm.
- Off Delay is a configurable period of time which the sensor must be inactive before the Bin Full condition is cleared.
- Action will configure the behavior of the system, when the bin becomes full, to immediately stop (default option) or to stop when the next product rejection is required.

See also

[Bin Full Output](#) ► Page 48

[Air Supply Sensor Input](#) ► Page 43

9.2.7 Bin Unlocked Input

This function can be used to define how the system responds when the Unlock Bin button is hit. It can be used in conjunction with a physical lock and key.

Options

There are four options associated with this function:

- Active Level
 - Select Active High or Active Low as appropriate.
- Reject Action determines system response when bin is unlocked:
 - Continue Reject
 - Inhibit Reject
 - Stop (conveyor will stop on detection of next contaminated product)
- Bin Open Time is the preset period of time:
 - The expected unlocked time (required to remove items from the bin).

A fault alarm will be raised if this time is exceeded.

See also

- [Bin Unlocked Input ▶ Page 57](#)
- [Bin Unlock Output ▶ Page 49](#)
- [Input and Output Configuration ▶ Page 40](#)

9.2.8 Conveyor Running Input

Conveyor Running is a mandatory input when the conveyor Start/Stop Control is defined as External.

The function informs the control system that the conveyor is running. Typically, this would be connected to the Run signal from a Variable Frequency Drive (VFD) or Inverter.

Options

There is one option associated with this function:

- Start/Stop Tolerance
 - The tolerance allows the user, for example, to take into account the time required for a conveyor to get up to speed or to stop.

9.2.9 Conveyor Start Input

This input function provides the means to start the conveyor from either user controls or external system. It is required when the conveyor Start/Stop Control is configured to one of the following:

- System IO
- UI and System IO

Options

There are no options associated with this function.

See also

- [Conveyor Stop Input ▶ Page 44](#)

9.2.10 Conveyor Stop Input

This input function provides the means to stop the conveyor from either user controls or external system. It is required when the conveyor Start/Stop Control is configured to one of the following:

- System IO
- UI and System IO

Options

There are no options associated with this function.

See also

- [Conveyor Start Input ▶ Page 44](#)

9.2.11 Downstream Blocked Input

This input function can be driven by equipment downstream of the metal detector in the event that the downstream equipment is not able to operate, requiring the conveyor to stop.

The Downstream Blocked input triggers a Downstream Blocked alarm. The alarm does not require acknowledgment.

Once the condition is removed for the downstream equipment, the operator will be able to reset and clear the alarm, allowing the conveyor to be restarted.

Options

There are no options associated with this function.

See also

[Upstream Starved Input](#) ► Page 47

9.2.12 Pack Sensor Input

This function allows the counting of discrete packs (products), and allows the option of synchronized rejection.

Options

There are two options associated with this function:

- Pack Sensor Is After Head
 - If enabled, this will indicate the pack sensor has been fitted downstream of the metal detector.
- Pack to Head Distance
 - This value is the measured distance from the pack sensor to the edge of the aperture and is used to provide a starting value for reject calculations, which can be refined later.
 - Changing this setting will not affect existing products.

See also

[Configure the Pack Sensor](#) ► Page 56

[Synchronized Reject Setup](#) ► Page 63

9.2.13 Reject Check Sensor Input

The reject check feature is a due diligence input, available on all PlusLine products and systems that have the PlusLine Protection Pack installed.

The sensor confirms that a product that should have been rejected does not leave the conveyor outfeed and works in conjunction with the pack sensor and reject product sensor.

Options

There are four options associated with this function:

- Window Offset
 - Distance or time window between Pack Sensor and Reject Check Sensor measured in ms or pulses.
- Window Extension
 - Extends the Pack Width, increasing the tolerance to compensate for any movement of products.
- Auto Calibrate
 - A setup wizard that calculates the Window Offset when a sample product is passed through the system.
- Consecutive Packs Before Warning
 - The number of unexpected packs at the Reject Check Sensor before the alarm is tripped.
 - The number of expected packs that are missing before the alarm is tripped (counted separately).

See also

[Reject Check Sensor](#) ► Page 59

9.2.14 Reject Operation Sensor Input

The Reject Operation Sensor input function performs a similar operation to the Rejected Product Sensor.

The Reject Operation Sensor would typically be used when the product being rejected is not a discrete product and cannot be confirmed as having entered the reject bin.

Instead, the Reject Operation Sensor confirms that the reject mechanism has been activated.

Options

There are three options associated with this function:

- Window Extension
 - Extension to the reject activation time to allow the reject confirmed signal to be received.
- On/Off Delay
 - A configurable period of time to filter spurious input signals from the Reject Operation Sensor.
- Reject Confirmation Enabled
 - Check box used to enable reject confirmation.
 - Can also be activated in System > > Detection and Rejection > Reject Confirmation Enabled

See also

- [Rejected Product Sensor Input ▶ Page 46](#)
- [Detection and Rejection Page ▶ Page 29](#)
- [Reject Output ▶ Page 51](#)

9.2.15 Rejected Product Sensor Input

The Reject Operation Sensor input function performs a similar operation to the Rejected Product Sensor.

The Rejected Product Sensor input function performs a similar operation to the Reject Operation Sensor.

The Reject Product Sensor would typically be used when the product being rejected is a discrete product and can be confirmed as having entered the reject bin.

Options

There are three options associated with this function:

- Window Extension
 - Rejected Product Sensor - The additional time required from when the reject device has removed a product from the conveyor (for example an actuator fully extended) and the product triggering the reject sensor.
- On/Off Delay
 - A configurable period of time to filter spurious input signals from the Rejected Product Sensor.
- Reject Confirmation Enabled
 - Check box used to enable reject confirmation.
 - Can also be activated in System > > Detection and Rejection > Reject Confirmation Enabled

See also

- [Reject Operation Sensor Input ▶ Page 45](#)
- [Detection and Rejection Page ▶ Page 29](#)
- [Reject Output ▶ Page 51](#)

9.2.16 Reset All Alarms Input

Each individual alarm will have its own default configuration (either latchable or not latchable).

Latchable alarms remain active even when the underlying cause has cleared. Such alarms will remain latched indefinitely and must be reset explicitly. A latchable alarm can only be reset when the alarm input condition is removed. Once a latchable alarm is reset, it is unlatched and becomes inactive.

Latchable alarms are useful for alerting users to, and maintaining, transient input conditions, that is conditions that come and go with no real duration, or conditions that are typically shorter than the time required for an operator to notice the condition.

By separating acknowledge and reset functionality, it is possible, for example, to provide open access to acknowledge functionality (for example via a push button) while limiting access to the reset function (for example via a reset key).

Options

There are no options associated with this function.

See also

- [Acknowledge All Alarms Input ▶ Page 42](#)
- [Acknowledge And Reset All Alarms Input ▶ Page 42](#)
- [Managing Alarms ▶ Page 84](#)

9.2.17 Safety Circuit Input

A Safety Circuit relay can monitor the status of safety devices such as emergency stops, door switches or drive related safety conditions (for example safe-torque off feature) via dual channel inputs.

The metal detector monitors the input and notifies the user by way of a Safety Device Operated alarm.

Options

There are no options associated with this function.

See also

- [List of Alarms ▶ Page 86](#)

9.2.18 Speed Sensor Input

A Speed Sensor enables the system to accurately determine system speeds by using pulse counts rather than fixed time duration. This improves accuracy of reject devices and reject confirmation of the product.

The speed is calculated by either an optical sensor or rotary encoder integrated into the conveyor system. The digital clock pulses created by these components are sent to the metal detector and provide information on conveyor speed and distance traveled.

Options

There are four options associated with this function:

- Conversion Method
 - Pulses/rev or Pulses/metre or Pulses/foot.
- Distance Encoder Ratio
 - The number of pulses required per unit of belt travel.
- Roller Diameter
 - Drive roller diameter.
- Belt Thickness
 - PU or modular belt thickness.

See also

- [Speed Settings ▶ Page 54](#)
- [Conveyor Running Input ▶ Page 44](#)

9.2.19 Upstream Starved Input

The Upstream Starved input function can be driven by equipment upstream of the metal detector. In the event that the upstream equipment is not able to run, it might be desirable to stop the conveyor.

The Upstream Starved input triggers an 'Upstream Starved' alarm, which does not require acknowledgment and is latching.

Once the condition is removed by the upstream equipment, the operator will be required to reset and clear the alarm, then to restart the conveyor.

This feature may be used in conjunction with Downstream Blocked.

Options

There are no options associated with this function.

See also

- [Downstream Blocked Input ▶ Page 45](#)

9.3 Managing Outputs

The availability of the following features will depend on the individual metal detector specification. Depending on the features selected, other associated features may be hidden.

External Computing Devices

External computing devices connected to the IO expansion outputs (where fitted) must be Limited Power Source and SELV circuit, and certified to the respective IEC/UL/CSA standards (i.e. IEC 60950-1, IEC 62368-1 or similar standards).

9.3.1 Setting up Outputs

Metal Detector	Access Level Required
All Metal Detectors	Administrator

The Outputs page shows all available outputs as tiles. Each tile contains the following information:

- The **Title** is the assigned function (default - Unused).
 - Select the tile that the external device you want to set up is connected to on the CPU I/O board.
- The **I/O board** where the function is located:
 - Main I/O = CPU I/O board.
 - Serial No = Expansion I/O board (optional).
- The **Active status** of the input.
 - Refer to Input/Output configuration for further details.

- 1 Navigate to **System > > System Integration > Outputs**.
- 2 Select required Tile.
- 3 Select the required **Function** from drop-down.
- 4 **Edit** the parameters as required.
 - ➔ The number of cells will vary depending on the device being set up.

Note The Active Level is set automatically, based on the chosen function. The box remains inactive and gray.

See also

- [Output Configuration ▶ Page 41](#)
- [Managing Outputs ▶ Page 48](#)

9.3.2 Unused

These are spare ports.

The location of all the ports is indicated:

- Main I/O (CPU I/O board) - 1 to 6.
- Where fitted "I/O expansion board serial Number" - 1 to 6.

9.3.3 Air Supply Failure Output

This output function is used in conjunction with the Air Supply Sensor input function. When an air fail condition is detected, the Air Supply Failure output can be used to notify users via a warning beacon or similar.

Options

There are no options associated with this function.

See also

- [Air Supply Sensor Input ▶ Page 43](#)

9.3.4 Bin Full Output

This output function is used in conjunction with the Bin Full Sensor input function. When a Bin Full condition is determined, the Bin Full output can be used to notify users via a warning beacon or similar.

Options

There are no options associated with this function.

See also

[Bin Full Sensor Input](#) ▶ Page 43

9.3.5 Bin Unlock Output

This output function controls the bin's electro mechanical lock. When enabled, the Unlock Bin button will be displayed on the UI. When this button is selected, the lock is opened for the prescribed period of time.

Options

There is one option associated with this function:

- Bin Unlock Time
 - The duration that the bin is unlocked.

See also

[Bin Unlocked Input](#) ▶ Page 44

[Bin Unlocked Output](#) ▶ Page 49

[Bin Unlock Function](#) ▶ Page 56

9.3.6 Bin Unlocked Output

This function is a configurable relay status output that activates immediately when the bin is determined to be unlocked via the Bin Unlocked Input.

Options

There are three options associated with this function:

- Active Level
 - Relates to the relay state, normally - Closed or Open.
- On Delay
 - Is a configurable period of time between activating the Unlock Bin button and the bin being unlocked.
- Off Delay
 - Is a configurable period of time from when the bin closed signal is activated and the bin is closed.

See also

[Bin Unlock Output](#) ▶ Page 49

[Bin Unlock Function](#) ▶ Page 56

9.3.7 Conveyor Drive Output

This output is used in Software Centric configurations. It is the output that connects directly to the inverter or Variable Frequency Drive (VFD).

Options

There are no options associated with this function.

9.3.8 Current Alarms Output

This function can be used to indicate whether there are any current alarms present. It is a digital output that is activated if any alarms are retained (that are active and/or unacknowledged), irrespective of their severity.

Options

There are no options associated with this function.

See also

[Current Fault Alarms and Higher Output](#) ▶ Page 50

[Current Warning Alarms Output](#) ▶ Page 50

[Current Warning Alarms and Higher Output](#) ▶ Page 50

[Managing Alarms](#) ▶ Page 84

9.3.9 Current Fault Alarms and Higher Output

This output function can be used to indicate whether there are any high severity current alarms present (that is faults or critical faults). It is a digital output, activated if any high severity alarms are retained (that are active and/or unacknowledged).

Options

There are no options associated with this function.

See also

- [Current Alarms Output ▶ Page 49](#)
- [Current Warning Alarms Output ▶ Page 50](#)
- [Current Warning Alarms and Higher Output ▶ Page 50](#)
- [Managing Alarms ▶ Page 84](#)

9.3.10 Current Warning Alarms Output

This output can be used to indicate whether there are any medium severity current alarms present (warnings). It is a digital output, activated if any warnings are retained (warnings that are active and/or unacknowledged).

Options

There are no options associated with this function.

See also

- [Current Alarms Output ▶ Page 49](#)
- [Current Fault Alarms and Higher Output ▶ Page 50](#)
- [Current Warning Alarms and Higher Output ▶ Page 50](#)
- [Managing Alarms ▶ Page 84](#)

9.3.11 Current Warning Alarms and Higher Output

This output function can be used to indicate whether there are any current alarms present with medium severity or higher (that is warnings, faults or critical faults). It is a digital output that is activated if any warning or higher severity alarms are retained (that are active and/or unacknowledged).

Options

There are no options associated with this function.

See also

- [Current Alarms Output ▶ Page 49](#)
- [Current Fault Alarms and Higher Output ▶ Page 50](#)
- [Current Warning Alarms Output ▶ Page 50](#)
- [Managing Alarms ▶ Page 84](#)

9.3.12 Detect Output

This function provides a digital output that is activated when the metal detector sees a detection event (irrespective of whether the conveyor is running or not). This can be used to drive a warning beacon and/or audible alarm.

Options

There are two options associated with this function:

- Latch Output
 - Enabling the option latches the physical output. An information level alarm is generated which does not require acknowledgment but does require reset for the output to be deactivated.
- Minimum Output On Time
 - The output will be active for at least the Minimum Output On Time value.

9.3.13 Good Pack/Product Output

This output function will activate the Reject Status Indicator on the Home page giving a visual indication of the product inspection result.

Options

There is one option associated with this function:

- Active Level
 - Select Open or Closed as appropriate.

See also

[Home Page](#) ► Page 16

9.3.14 Latched Alarms Output

This output function provides a digital output that is active if there are any currently latched alarms, that is alarms that do not have an active input signal but are latched active.

Can be used to indicate if the alarms reset button can be pressed for example.

Options

There are no options associated with this function.

9.3.15 Performance Test Due Output

This function provides a digital output that is active when a Performance Test is due. The output can only be deactivated by clearing the Performance Test Due condition.

Options

There are no options associated with this function.

See also

[Performance Test Overdue Output](#) ► Page 51

9.3.16 Performance Test Overdue Output

This function provides a digital output that is active when a Performance Test is overdue. The output can only be deactivated by clearing and resetting the Performance Test Overdue condition.

Options

There are no options associated with this function.

See also

[Performance Test Due Output](#) ► Page 51

9.3.17 ProdX Disconnected Output

This output function provides a digital output that will activate when the system disconnects from ProdX.

Note ProdX has to be enabled in 'ProdX Settings' for this output to work.

Options

There are no options associated with this function.

See also

[ProdX](#) ► Page 59

9.3.18 Reject Output

This output function is a digital output that activates when the product is rejected. The Reject output should not be used for Stop on Detect applications - instead, the Stop on Detection output should be used.

Options

There are two options associated with this function:

- Reject Mechanism configures the type of reject mechanism installed on the system:

- Air-blast
- Custom
- Diverter
- Double Bag
- Flap
- Overhead Sweep
- Punch
- Pusher
- Retracting Band
- Head to Reject Distance
 - The distance between the side of the head facing the reject and the Reject Mechanism.

Note It is important to configure the reject mechanism correctly to ensure the correct reject timing. For example, the behavior for a retracting band reject will be different to an air-blast.

See also

- [Stop On Detection Output ▶ Page 52](#)
- [Reject Operation Sensor Input ▶ Page 45](#)
- [Rejected Product Sensor Input ▶ Page 46](#)
- [Detect Output ▶ Page 50](#)

9.3.19 Reject Not Confirmed Output

The Reject Not Confirmed output is a digital output. When Active, it indicates one of the following:

- The rejected product did not enter the reject bin.
- The Reject Mechanism did not operate.
- The Reject Check Sensor detects a product identified for rejection.

Options

There are no options associated with this function.

See also

- [Reject Operation Sensor Input ▶ Page 45](#)
- [Rejected Product Sensor Input ▶ Page 46](#)
- [Reject Check Sensor ▶ Page 59](#)

9.3.20 Stop On Detection Output

This is a digital output that activates when a metal detection has caused a Stop On Detect conveyor to stop.

The Stop on Detection output should not be used for automated Reject applications, instead, the Reject output should be used.

Options

There are no options associated with this function.

See also

- [Reject Output ▶ Page 51](#)

9.3.21 System Healthy Output

The System Healthy Output (when configured) is used to monitor the operational status of the Metal Detector as follows:

- When the system is operational, the output is healthy.

Note This output function can trigger a visual and/or audible alarm when an event occurs.

The following table gives examples of alarm conditions and their effect on the System Healthy Output.

Input Condition	Effect on System Healthy
Air Supply Failure	Yes

Input Condition	Effect on System Healthy
Battery Fault	Yes
Bin Full	Yes
Contaminated Pack Not Rejected	Yes
Conveyor Running Fault	Yes
Conveyor Running Not Confirmed	Yes
Detection Event	Yes
Downstream Blocked	Yes
Pack Rate Too High	Yes
Pack Sensor Blocked	Yes
Pack Space Warning	Yes
Pack(s) Missing at Reject Check Sensor	Yes
Performance Test Due	Yes
Performance Test Overdue	Yes
Product Needs Auto Setup	Yes
Product Not Setup	Yes
Reject Activation Inhibited By Unlocked Bin	Yes
Reject Bin Unlocked	Yes
Reject Check Before Reject Scheduled	Yes
Reject Check Not Setup	Yes
Reject Check Sensor Blocked	Yes
Reject Operation Not Confirmed	Yes
Reject Operation Sensor Fault	Yes
Rejected Product Not Confirmed	Yes
Rejected Product Sensor Fault	Yes
Safety Device Operated	Yes
Speed Sensor Fault	Yes
System Stopped By Unlocked Bin	Yes
Unexpected Pack(s) at Reject Check Sensor	Yes
Upstream Starved	Yes
Conveyor Stopped	No Effect
Reject Inhibited	No Effect
Detection Event	No Effect
Fail Safe Rejection	No Effect
Reject Bin Unlocked	No Effect
Fail Safe Detect	No Effect
ProdX Server Unavailable	No Effect

Options

There are no options associated with this function.

Note A visual and/or audible alarm will be given where there is an effect on System Healthy.

9.3.22 System Stop Output

This is a required output for systems where the Start/Stop Control is set to None or External.

Its purpose is to notify systems where the metal detector has no knowledge of the conveyor state and that the system should now be stopped, for example, after a high severity alarm has occurred.

Options

There are no options associated with this function.

See also

[The Start/Stop Control](#) ► Page 55

9.3.23 Unacknowledged Alarms Output

This output activates when there are unacknowledged alarms present.

Options

There are no options associated with this function.

9.4 Speed Settings

See also

[Speed Sensor Input](#) ► Page 47

9.4.1 Speed Sensor

The Speed Sensor is either an optical sensor or a rotary encoder configured on the conveyor system to create digital clock pulses.

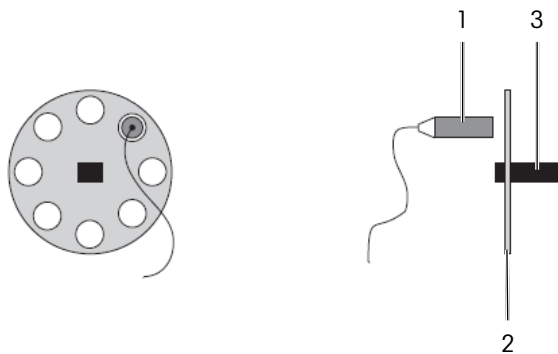
These pulses are sent to the metal detector to provide information on conveyor speed and distance traveled.

You must use a Speed Sensor on variable speed systems.

A Speed Sensor is an optional component on METTLER TOLEDO metal detection conveyor systems.

The M30 R-Series Installation Manual provides details of the connections to the CPU I/O board and technical specification requirements.

The following figure shows a typical Speed Sensor:



Speed sensor configuration

1	Optical sensor	2	Disc
3	Drive shaft		

9.4.2 Setting up Conveyor Speed

Metal Detector	Access Level Required
All Metal Detectors	Administrator

To configure speed settings:

- 1 Navigate to **System > > Speed Settings**.
- 2 From the **Speed Setting** drop-down list, select **Variable** or **Fixed**.
- 3 Select the appropriate option and enter the required settings:

Settings

If Variable:

- Select Maximum Speed:
 - Either manually enter required speed value.
 - Or select from the pre-configured Max, Default or Min values.

Note The variable speed option should only be used when a speed sensor is fitted to the conveyor.

If Fixed:

- Select Maximum Speed:
 - Either manually enter required speed value.
 - Or select Auto Speed to run the wizard to determine the speed.

Note The Auto Speed wizard will ask you to start the conveyor and pass a metal sample. Follow the on screen instructions.

9.5 Conveyor Start/Stop Control Options

Metal Detector	Access Level Required
All Metal Detectors	Administrator

The Start/Stop Control feature allows you to configure how the conveyor starts and stops. See table below for available options.

The Start Guard Enabled check box, provides an additional safety step. When enabled, a confirmation step is added when starting the conveyor.

The Start/Stop Control options require certain input or output functions to be configured:

Option	Description	Required input(s)	Required output(s)
None	No start/stop control. The product moves along the conveyor using gravity, for example via rollers.	None. Safety Circuit is optional.	System Stop
External	The conveyor is not controlled via the software. Either a control circuit or an external system is used.	Conveyor Running Safety Circuit is optional.	System Stop
System IO	The conveyor is controlled via the software, using the system IO and physical start/stop buttons. No start/stop buttons are available on the UI.	Conveyor Start Conveyor Stop Safety Circuit	Conveyor Drive
UI	The conveyor is controlled via the software, using the system IO and start/stop buttons on the UI. Physical start/stop buttons are not used.	Safety Circuit	Conveyor Drive
UI and System IO	The conveyor is controlled via the software, using the system IO. Physical start/stop buttons and UI buttons are both available.	Conveyor Start Conveyor Stop Safety Circuit	Conveyor Drive

9.5.1 The Start/Stop Control

Metal Detector	Access Level Required
All Metal Detectors	Administrator

To configure the Start/Stop Control:

- 1 Navigate to **System > > System Integration > Conveyor**.
- 2 Select **Start/Stop Control**.
- 3 Select required Option.
 - ➔ **Note** If an associated required input or output has not been configured, a "metal detector has stopped working" alarm is given detailing which input/output requires configuring.
- 4 To activate **Start Guard Enabled**, select check box.
 - ➔ When the conveyor start button is selected, a choice of **Start Conveyor** or **Cancel** is displayed to confirm the selection.

See also

 Alarm Icons and their Meaning ► Page 84

9.6 Configure the Pack Sensor

Metal Detector	Access Level Required
All Metal Detectors	Administrator

The Pack Sensor (usually a light beam sensor) is normally positioned on the conveyor system at the infeed side of the metal detector. However, if required by the application, it is also possible to mount the Pack Sensor after the metal detector.

To configure the Pack Sensor:

- 1 Navigate to **System > > System Integration > Inputs**.
- 2 Select the "unused" Input to which the sensor has been connected.
- 3 From the **Function** drop-down, select **Pack Sensor**.
- 4 From the **Active Level** drop-down, select **Active High** or **Active Low** as appropriate.
- 5 Enter the **Pack to Head Distance**.
- 6 Select **Pack Sensor Is After Head** if required.

For a description of the Pack Sensor, and how it interacts with the other sensors on the system, refer to your GC system Operator manual.

See also

[Pack Sensor Input](#) ► Page 45

9.7 Machine Status (PackML State)

Metal Detector	Access Level Required
All Metal Detectors	Administrator

You can display the machine state (PackML) on the UI Home page.

To do this:

- 1 Navigate to **System > > System Integration > PackML**.
- 2 Select **Show PackML State**.

The machine state will then appear to the bottom right of the bargraph.

Note If the Unlock Bin feature is active, PackML messages will be hidden behind the Unlock Bin button.

9.8 Bin Unlock Function

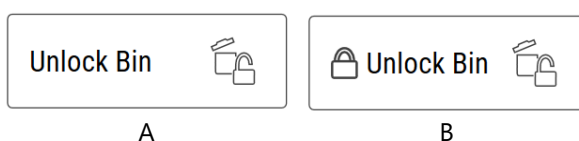
9.8.1 Unlocking the Bin

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R StandardLine - M31R with PlusLine Protection Pack	All Users

The Bin Unlock feature prevents products from being removed from the reject bin by an unauthorized person. It can also prevent the metal detector from operating for sustained periods without the reject bin being securely closed.

When activated, the Unlock Bin button (A) will be shown on most pages of the user interface.

When no one is logged in a closed padlock appears (B) indicating the feature is locked and sign-in will be required.



On tapping the Unlock Bin button it grays out:

- If the bin is not opened, the button returns active after the preset Bin Open Time has elapsed.
- When the bin is opened, a second Bin Open Time count starts related to the open bin:
 - During this period, a bin open Warning Alarm is raised until the bin is closed.
 - If the bin remains open beyond the preset Bin Open Time, a Fault Alarm is raised, which must be reset.

Note Any further action required will depend on the Unlock Bin settings.

Note The two "Bin Open" time counts are independent, the Unlock Bin button will re-activate while the Open Bin count continues it's countdown.

9.8.2 Bin Unlocked Input

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R StandardLine - M31R with PlusLine Protection Pack	Administrator

Settings

Configure the Reject Action on Bin Unlock:

- Continue Reject
 - Contaminated products will continue to be ejected from the conveyor while the bin is open.
- Reject Inhibit
 - Contaminated products will not be ejected but continue along the conveyor while the bin is not secured closed.
- Stop on Detection
 - The next contaminated product detected will stop the conveyor. The system will require resetting after the bin is secured closed.

Configure the alarm state (Enable Fault Alarm) on Bin Unlock:

- Bin open Warning
 - Is shown when the bin is opened and clears when the bin is securely closed.
 - The warning alarm can be acknowledged during this period.
- Bin open Fault
 - Is shown when the specified Bin Open Time is exceeded.
 - The fault alarm requires resetting.

Setup Bin Unlocked Input

- 1 Navigate to **System > > System Integration > Inputs**.
- 2 Select the "unused" Input to which the sensor has been connected.
- 3 Set Function to **Bin Unlocked**.
 - ➡ Additional properties are then shown.
- 4 Select appropriate **Active Level** from drop-down list.
 - ➡ This relates to the relay behavior - normally open or normally closed.
- 5 Enable or disable **Alarm Enabled** as required.
- 6 Select appropriate **Reject Action**.
- 7 Set **Bin Open Time** as required.

See also

- 🔗 Set-up the Reject Check Sensor ▶ Page 60
- 🔗 Bin Unlock Outputs ▶ Page 58

9.8.3 Bin Unlock Outputs

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Protection Pack	

9.8.3.1 Bin Unlock

Setup Bin Unlock Output

- 1 Navigate to **System** > > **System Integration** > **Outputs**.
- 2 Select the "unused" Input to which the sensor has been connected.
- 3 Set Function to **Bin Unlock**.
- 4 Select appropriate **Active Level** from drop-down list.
 - ➔ This relates to the relay behavior - normally open or normally closed.
- 5 Set **Bin Unlock Time** as required.
 - ➔ When activating **Bin Unlock** the lock will be de-activated (unlocked) for the specified bin unlock time period before re-activating (locked).

Note When the Output Function is set to Bin Unlock, an Unlock Bin button appears below the bargraph. This button will appear on most pages when navigating around the User Interface.

When no one is logged in a closed padlock appears (B) indicating the feature is locked and sign-in will be required.



See also

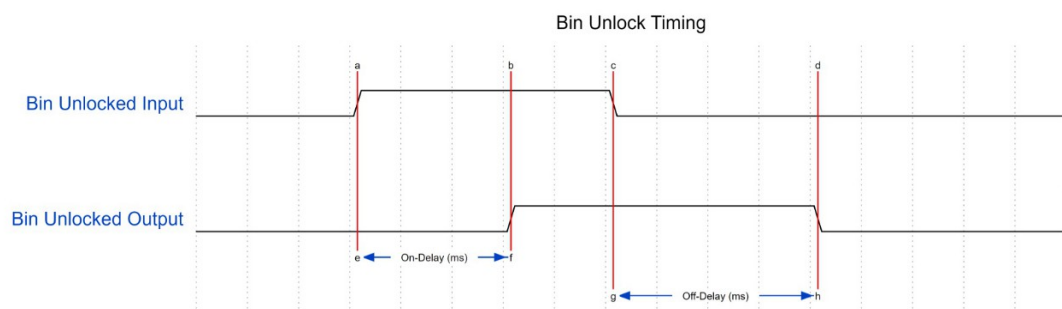
[Bin Unlock Output](#) ▶ Page 49

9.8.3.2 Bin Unlocked

This output is only available if the Bin Unlock input is configured.

Setup Bin Unlocked

- 1 Navigate to **System** > > **System Integration** > **Outputs**.
- 2 Select the "unused" Output to which the sensor has been connected.
- 3 Set Function to **Bin Unlocked**.
- 4 Select appropriate **Active Level** from drop-down list.
 - ➔ This relates to the relay behavior - normally open or normally closed.
- 5 Set **On Delay** as required.
- 6 Set **Off Delay** as required.



See also

[Bin Unlocked Input](#) ▶ Page 57

[Bin Unlocked Output](#) ▶ Page 49

9.9 ProdX

Metal Detector	Access Level Required
All Metal Detectors	Administrator

It is recommended that ProdX is set up via the ProdX software.

To activate ProdX:

- 1 First set up the metal detector in the **ProdX** software, prior to activating **ProdX** in the metal detector.
 - ➔ This auto-fills the **ProdX** Server settings in the metal detector.
- 2 Navigate to **System > > ProdX Settings**.
- 3 Select the **ProdX Enabled** box.
 - ➔ A small X appears top right of the bargraph, signifying **ProdX** is active:
 - ➔ Blue X signifies **ProdX** is enabled and connected.
 - ➔ Gray X signifies **ProdX** is enabled and disconnected.
- 4 Select the appropriate **Disconnect Action**, see table below.
 - ➔ This is the action required should the connection between the **ProdX** software and the metal detector be lost.

Disconnect Action

Action	Result
Do Nothing	Event is logged by the metal detector only.
Warning Level Alarm (2532)	ProdX Disconnected Warning. ProdX Server connection has been lost, please check network connectivity. Amber warning - No Action
Fault Level Alarm (2531)	ProdX Disconnected Fault. ProdX Server connection has been lost, please check network connectivity. Red warning - Stop

9.9.1 Manual ProdX Setup

Metal Detector	Access Level Required
All Metal Detectors	Administrator

- 1 Navigate to **System > > ProdX Settings**.
- 2 Enter the required settings:
 - ➔ **Device Server Port**
 - ➔ **ProdX Server IP**
 - ➔ **ProdX Server Port**
- 3 Select required **Disconnect Action**.
- 4 Select **ProdX Enabled** check box to activate **ProdX**.

9.10 Reject Check Sensor

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R StandardLine - M31R with PlusLine Protection Pack	Administrator

To switch reject confirmation on:

- 1 Navigate to **System > > Detection and Rejection**.
- 2 Select the **Reject Confirmation Enabled** box.
- 3 For full due diligence functionality, ensure the **Unsynchronized Detection Events Enabled** box is also ticked.

9.10.1 Set-up the Reject Check Sensor

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Protection Pack	

Auto Calibrate Reject Check Sensor

- 1 Navigate to **System > > System Integration > Inputs**.
- 2 Select the "unused" Input to which the sensor has been connected.
- 3 In **Function**, select **Reject Check Sensor**.
- 4 **Accept**.

Note Unsynchronized detection events are unexpected signals such as:

- Missing or extra product
- Failed reject

9.10.1.1 Reject Check Sensor Main I/O Page Description

- Inactive (High or Low) indicator:
 - Linked to Active Level selection.
 - Blue signifies the Active level (High or Low) read from the sensor. When the sensor is triggered, the output will switch to the opposite (Low or High), corresponding to the PNP/NPN setting for that input.
 - Gray signifies the Inactive level (High or Low) read from the sensor. When the sensor is triggered, the output will switch to the opposite (Low or High), corresponding to the PNP/NPN setting for that input.
- Active Level - Active Low or Active High:
 - Trigger signal either low or high voltage output.
- Window Offset:
 - Distance or time window between Pack Sensor and Reject Check Sensor measured in ms or pulses.
 - Set during Auto Calibrate of the Reject Check Sensor.
- Window Extension:
 - Default 0
 - Extension of the pack width window.
 - Useful where a product may move on the conveyor as it is transported.
- Consecutive Packs Before Warning:
 - Default 1.
 - The number of unexpected packs at the Reject Check Sensor before the alarm is tripped.
 - For example, an object falling on the conveyor downstream of the pack sensor.

9.10.1.2 Reject Check Sensor Main I/O Page Set-up

- 1 Select appropriate **Active Level** from drop-down list.
- 2 Adjust **Window Extension** value as required.
- 3 Adjust **Consecutive Packs Before Warning** value as required.
 - ➔ **Note** For full due diligence, **Unsynchronized Detection Events Enabled** must be enabled.
- 4 **Start** the conveyor.
- 5 Select **Auto Calibrate**.
- 6 Pass a representative product sample through the metal detector aperture.
- 7 On successful completion, a message "**Routine complete - Accept changes**" is shown.
 - ➔ Select **Accept** if you are happy with the result.
 - ➔ Select **Cancel** if not and re-run **Auto Calibrate**.

Note The **Window Offset** value has been updated according to the accepted result.

See also

 Reject Check Sensor Input ► Page 45

9.10.2 Reject Check Error Messages

The following alarms may be seen:

- Contaminated Pack Not Rejected - alarm code 2050.
- Reject Check Sensor Blocked - alarm code 2051.
- Pack(s) Missing at Reject Check Sensor - alarm code 2052.
- Unexpected Pack(s) at Reject Check Sensor - alarm code 2053.

See also

[🔗](#) List of Alarms ► Page 88

10 Managing Products

All products that require checking must be defined in the metal detector prior to inspection runs.

10.1 Preconditions / Considerations



Important Safety Message

Speed Settings should only be adjusted on initial setup - or for a major reconfiguration of the metal detection system.

Pre-existing product settings may no longer be valid, leading to contaminated products not being detected/rejected and a Product Not Setup alarm.

- Speed Settings must be set before setting up a product.
 - **Note** Changing the Speed Settings will result in a Product Not Setup alarm.
- The Pack Sensor, where fitted, must be configured correctly.
- For optimal Sensitivity, Phase and Mode settings (calculated) for the product, it is recommended that the Auto Setup feature is used. A representative product sample can then be passed through the aperture of the metal detector.
- During the Auto Setup, the reject mechanism is usually disabled to make sure that no products are unnecessarily rejected during a product setup.
- If you are working in a live production environment, you can disable Reject Inhibit (in System/Auto Setup Settings) to make sure that contaminated product is rejected from the production line. However, this is only possible if you are confident that the active product settings are valid, and correctly set up.

When using Auto Setup:

- Auto Setup provides on-screen instructions when setting up a product.
- The system needs to be running.
- If the sample products are discrete products, they must be spaced.
- Mode Selection contains a list of operating frequencies or modes (depending on detector type) available. You can select an individual frequency or mode, or use the Auto Setup function to ensure the optimum settings are achieved:
 - When using , Auto Setup the machine selects the mid-range frequency and undertakes a pass. It then calculates the results. If out of range, for that frequency, it selects the next likely frequency and requests a second pass. This is repeated until the optimum frequency or mode setting for the machine is determined for that product.
- For M31R, M33R and M33RB systems, product setup can be done manually or by using the Auto Setup feature.
- For the M34R system, Auto Setup must be performed at least once for each product.

See also

- [Configure the Pack Sensor ▶ Page 56](#)
- [Speed Sensor ▶ Page 54](#)
- [Setting up Conveyor Speed ▶ Page 54](#)

10.2 Setting up Products

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

Prerequisites

- A representative sample selection for the product under inspection is required.
- Approved metal test samples are required.

10.2.1 Setting up a Product for the First Time

10.2.1.1 Product Setup

If you are setting up a product for the first time:

- 1 Navigate to **Products > Product List**.
- 2 Select **New**.
- 3 Enter **Product** name and **Code**.
 - ➔ The product **Code** is usually optional. However, in **Products >> Product Rules** the **Engineer** or the **Administrator** can make the **Code** mandatory by enabling **Mandatory Product Code Enabled**.
 - ➔ The product name, and when used product code, must be unique for each product.
- 4 Select **Next**.
 - ➔ If the product is new, the message **Do the setup now?** appears, with the buttons **Continue**, **Save and Exit**.
 - ➔ If you create a new product, but do not complete the setup procedure, it will appear in the product list with the text **Requires Setup**.
- 5 Select **Continue**.
 - ➔ The **Start Auto Setup** screen appears, in the **Mode Selection** field, the default value is **Auto**. However, if the **Speed Settings** of the system are too slow, **Auto** will not be available.
- 6 Ensure the system is operational.
- 7 Select the **Start** button to commence the **Auto Setup** sequence.
- 8 Follow the on screen instructions, pass a single product, representative of your product range, through the metal detector aperture.
- 9 Once you have passed a representative sample of product - Select **Done**.
- 10 Follow the on screen instructions, pass a range of products, representative of your product range, through the metal detector aperture.
 - ➔ The metal detector will calculate the optimum **Mode**, **Sensitivity** and **Phase** for the range of products inspected.
- 11 Once you have passed a representative sample of product - Select **Done**.
- 12 If required select **Repeat** to re-run **Auto Setup**, or select **Next** to continue with product setup.

10.2.1.2 Adjust Inspection Settings

The inspection settings are now active, the Mode, Sensitivity and Phase can be adjusted - fine tuned if required.

- The Advanced Settings option provides additional levels of Sensitivity and Phase.

To adjust settings:

- 1 Adjust a single value (**Mode**, **Sensitivity** or **Phase**) at a time.
- 2 Pass a representative product sample through the metal detector aperture.
- 3 Observe the Bargraph:
 - ➔ Aim to have the first five green bars indicated on the bargraph when the representative product sample passes through the metal detector.
 - ➔ Adjust the values until you are satisfied with the output.
- 4 Select **Next**:
 - ➔ If a pack sensor is configured in System Inputs, the setup will proceed to **Synchronized Reject Setup**.
 - ➔ If a pack sensor is not configured in System Inputs, the setup will proceed to Final Adjustments.

See also

 Final Adjustments ► Page 64

10.2.1.3 Synchronized Reject Setup

Note If there is no pack sensor, the Synchronized Reject option is not available.

- 1 Select **Yes** if you are using Synchronized Reject or **No** if you are not.

- ➔ By selecting **No**, set up continues, go to Final Adjustments.
 - ➔ By selecting **Yes** the options that follow depend on the Reject Mechanism on the conveyor system (set in **Function/Reject** in **Outputs**).
- 2 Pass samples, according to the on screen instructions.
 - 3 You can opt to re-run for **Synchronized Reject Setup**, or
 - 4 Select **Next**.

See also

🔗 Final Adjustments ▶ Page 64

10.2.1.4 Final Adjustments

For discrete products, it is recommended that the Reject Settings are adjusted until a reliable reject is achieved:

- Detection to Alarm - Travel time or distance between the Pack Sensor and activation of alarm.
- Detection to Rejection - Travel time or distance between the Pack Sensor and reject device
- Reject On Extension - Travel time or distance for reject mechanism.

- 1 If you do not wish to adjust values, select **Next**, else:
- 2 Adjust a single value (**Detection to Rejection**, **Reject On Extension**, **Pack to Reject** or **Pack Sensor Blocked Time**).
- 3 Pass a representative product sample with a metal test sample along the conveyor a number of times
 - ➔ Change the position of the metal test sample in the product each time.
- 4 Observe device to determine best setting.
- 5 Select **Next**.
- 6 Select **Save**.
 - ➔ The **Product List** page appears with the new product set as **Active**.



NOTICE

If critical metal detector settings are changed, or sensors added/replaced, existing products will require reconfiguring to ensure Metal Contamination continues to be detected.

For example, if a Speed Sensor is fitted or replaced, or Pack Sensor changed for synchronized reject products.

10.2.2 Creating a New Product from an Existing Product

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

- 1 Navigate to **Products > Product List**.
- 2 Select required product to copy.
- 3 Select **Copy** from pop up.
 - ➔ **Result** A copy of the selected product is generated.
- 4 Select the newly created **Copy**.
- 5 Select **Setup Wizard**.
- 6 Follow the on screen prompts to rename and configure the new product.

10.3 Viewing and Changing Products

Metal Detector	Access Level Required
All Metal Detectors	Operator, Supervisor, Engineer or Administrator

In the Changeover and Product List pages, you can search for a specific product by pressing the filter icon , Search and Sort By fields are displayed. Enter the search string, then filter using options in the Sort By field.

Changing the Active product

- 1 Navigate to **Products > Changeover**.
- 2 Select the **Product** you wish to make **Active**.
- 3 Select **Changeover**.
 - ➔ The name of the **Active** product is displayed in the large icon on the top-right of the page.
 - ➔ In the Product List, the word **Active** appears on the product icon of the current product.

Note If you have a higher access level than Operator, you can also Changeover by searching the Product List and selecting a product. Then select Changeover from the pop up menu.

10.4 Editing Products and Product Settings

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator (Supervisor has some limited access to product settings)

For best results, METTLER TOLEDO recommend that you set up a product using the Auto Setup feature.

However, following Auto Setup you may still observe high levels of variation in the product signal on the bargraph resulting in false detections and rejections. If this happens, there are a number of options available to make improvements:

- 1 Navigate to **Products > Auto Setup**.
- 2 Run **Auto Setup** again with an increased number of products and more varied presentation.
- 3 Decrease the target bargraph level, in Set Signal Level for **Auto Setup**. Similarly, if the product signal is very consistent, the target bargraph can be increased to improve detection capability.
 - ➔ **Note** This is a **System** level change and will affect all future products.
- 4 Manually change the **Mode** and re-setup.
- 5 Manually adjust the product settings.

See also

-  Set Mode Selection Manually ▶ Page 65
-  Adjust Sensitivity and Phase Manually ▶ Page 66
-  Set Signal Level for Auto Setup ▶ Page 66

10.4.1 Set Mode Selection Manually

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator (Supervisor has some limited access to product settings)

Note Only the Active product can be edited.

- 1 Navigate to **Products > Auto Setup**.
- 2 In the **Mode Selection** menu, select a different **Mode**.
- 3 Re-run **Auto Setup**.
 - ➔ This ensures the selected Mode has been calibrated.

Note It is recommended that after adjusting the Mode value, where active, synchronized reject is checked. In some cases, synchronized reject may need re-adjusting.

Note For the M34R system, when manually changing the mode, the user will be warned that product auto setup is required. The conveyor start button will be grayed out until this is undertaken.

10.4.2 Adjust Sensitivity and Phase Manually

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator (Supervisor has some limited access to product settings)

Note Only the Active product can be edited.

- 1 Navigate to **Products > Product List**.
- 2 Activate the Product you wish to update:
 - ➔ Select the Product, then **Changeover**.
- 3 **Inspection Settings** can be accessed in any of the following ways:
 - ➔ Navigate to **Products > Product Settings >> Inspection Settings**
 - ➔ **Products > Product List**, select the product, then **Settings**.
- 4 Adjust the **Sensitivity** or **Phase** as required.
- 5 Pass a representative product sample through the metal detector aperture.
- 6 Observe the Bargraph:
 - ➔ Adjust the values until you are satisfied with the output.
- 7 Repeat for the other setting if required.

Note It is recommended that after manually adjusting the Sensitivity and more importantly Phase values, where active, synchronized reject is checked. In some cases, synchronized reject may need re-adjusting.

10.4.3 Set Signal Level for Auto Setup

Metal Detector	Access Level Required
All Metal Detectors	Administrator

To adjust a product's target bargraph level:

- 1 Navigate to **System > Auto Setup Settings**
- 2 Adjust value in **Signal Level for Auto Setup** as required.

10.5 Manually Adjust Non-Synchronized Reject Settings

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

Note Only the Active product can be edited.

Prerequisites

- A representative sample selection for the product under inspection is required.
- Approved metal test samples are required.

Note The Reject Settings page allows for the adjustment of the reject timings.

- 1 Navigate to **Products > Product Settings >> Reject Settings**.
- 2 Adjust the:
 - ➔ **Detection to Alarm** adjust value as required (no reject output configured).
 - ➔ **Detection to Rejection** adjust value as required (reject output configured).
 - ➔ **Reject On Extension** adjust value as required (reject output configured).
- 3 Pass an approved metal test sample with a product.
 - ➔ Ensure metal test sample is placed in various positions on the product.
- 4 Check the product is rejected as required.
 - ➔ Adjust values if necessary.

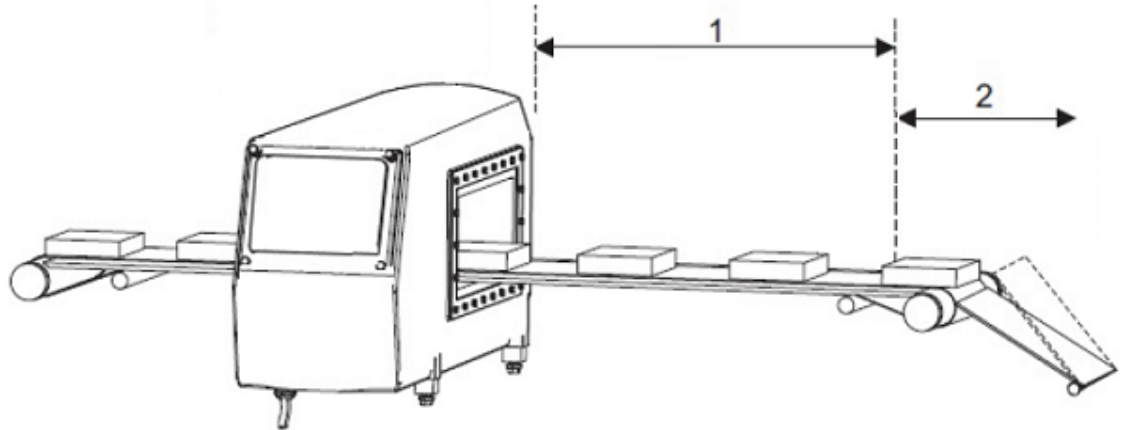
Settings

Reject Settings page

- Detection to Alarm - Travel time or distance between the Pack Sensor and activation of alarm.
- Detection to Rejection - Travel time or distance between the Pack Sensor and reject device
- Reject On Extension - Travel time or distance for reject mechanism.

Advanced page

- Pack Sensor Blocked Time (if fitted) - Period that the product blocks the Pack Sensor as it passes.



1. Detection to Alarm
2. Reject On Extension

10.6 Manually Adjust Synchronized Reject Settings

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

Note Only the Active product can be edited.

Prerequisites

- A Pack Sensor must be fitted to the system.
- Synchronized Reject must be set up for the product.
- A representative sample selection for the product under inspection is required.
- Approved metal test samples are required.

Note The Reject Settings page allows for the adjustment of the reject timings.

- 1 Navigate to **Products > Product List**.
- 2 Activate the Product you wish to update:
 - ➔ Select the Product, then **Changeover**.
- 3 Navigate to **Products > Product Settings >> Reject Settings**.
 - ➔ Here you can switch the **Synchronized Reject** check box on or off. A warning "Make sure that the settings are still valid" is given.
 - ➔ If the selected product has not been set up for **Synchronized Reject**, the **Synchronized Reject** check box will be grayed out.
- 4 Choose to **Set Up Synchronized Reject** or **Re-run Synchronized Reject Setup** as required, and follow the on screen prompts.
- 5 Else manually adjust the values.
- 6 Pass an approved metal test sample with a product.
- 7 Check the product is rejected as required.

➔ Adjust values if necessary.

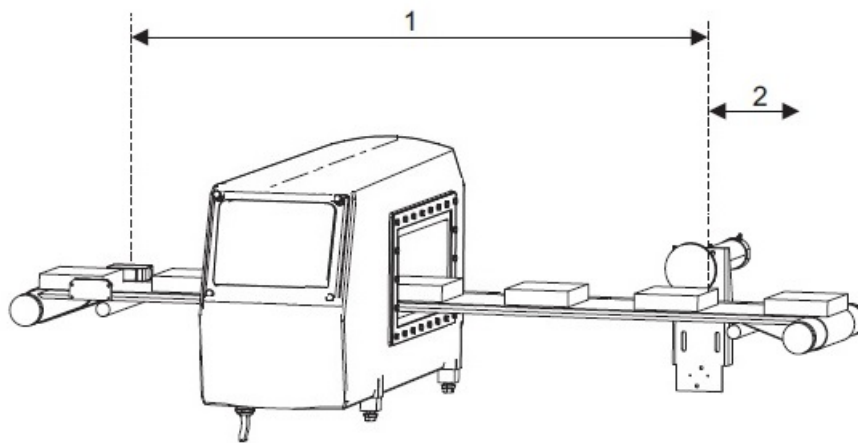
Settings

Reject Setting page

- Pack to Reject - Travel time or distance between the Pack Sensor and reject device
- Reject On Extension - Travel time or distance for reject mechanism.

Advanced page

- Pack Sensor Blocked Time (if fitted) - Period that the product blocks the Pack Sensor as it passes.
- Pack Length - The length of the product being passed.
- Pack Synchronization Tolerance - During setup, the detector calculates the position of the metal test sample in relation to the Pack Sensor. Adjusting the tolerance value changes the level of deviation $\pm$ of the calculated position. This may help attribute the detection event to the correct contaminated products.



1. Pack to RejectPack Sensor to reject device
2. Reject On Extension

See also

- [🔗 Final Adjustments ▶ Page 64](#)
- [🔗 Setting up a Product for the First Time ▶ Page 63](#)

10.7 Viewing and Resetting the Counters and Settings

Viewing

Metal Detector	Access Level Required
All Metal Detectors	All Users

To view the active product statistics:

- Navigate to **Statistics > Production**.

Resetting Counters

Metal Detector	Access Level Required
All Metal Detectors	Engineer or Administrator

To reset the active product statistics:

- 1 Navigate to **Statistics > Production**.
- 2 Tap **Reset**.

Configuring Settings

Metal Detector	Access Level Required
All Metal Detectors	Administrator

To Reset Counters on Product Changeover:

- 1 Navigate to **Statistics > Settings**.
- 2 Select **Reset Counters on Product Changeover** check box to activate.
- 3 De-select check box to de-activate.

11 Performance Testing

It is important that performance tests are carried out regularly, in accordance with your company's standards and quality assurance procedures. The period of performance tests, and test sample sizes to use during the test, are determined by your company's standards for metal detection.

The performance test confirms that the metal detector is working correctly, and confirms that it has been able to detect and reject contaminated products since the last performance test was done. The test applies to the currently selected Product. METTLER TOLEDO recommends that you do a new performance test after a product change or a change of settings.

The metal detector can be set up to monitor and undertake performance tests:

- If the performance test is overdue, a Warning message Performance Test Due appears on the Home page of the User Interface. The warning also appears on the Alarms page.

See also

 Managing Alarms ► Page 84

11.1 Test Sample Setup

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R StandardLine - M31R with PlusLine Productivity Pack	Administrator

Test sample setup defines the test samples that are to be used for a given product.

The choice of test samples should be defined in your company's approved standards and quality assurance procedures.

- 1 Navigate to **QA > Test Sample Setup**.
- 2 Select the required materials to be tested.
 - ➔ A tick is displayed in the enabled box, the test material cell will become active.
- 3 Select the test material cell, input the size of the test sample to be used.
- 4 **Accept**.

Note The test sample materials are color coded:

- Ferrous - **Red**
- Non-Ferrous - **Yellow**
- Stainless Steel - **Blue**

11.2 Setting up Performance Tests

11.2.1 StandardLine

Metal Detector	Access Level Required
StandardLine - M31R	Administrator

To enable a test:

- 1 Navigate to **QA > Test Setup**.
- 2 Select **Enabled**.
 - ➔ **Scheduling Mode** becomes active.
- 3 Adjust the settings as required.
 - Scheduling Mode:
 - Timed
 - Off
 - Due Interval:
 - Adjust the time interval between tests (maximum 24:00 hours).
 - Schedule Type:
 - Fixed Schedule - Test scheduled at a fixed rate, regardless of when they are run.

- Interval From Last - Test scheduled at an interval after the last test was run.
- Login Required
 - When enabled, login will be required to undertake the Performance Test.

Note When enabled, Scheduling Mode can be switched to Off, Test Setup will still show as enabled allowing tests to be run without scheduling.

11.2.2 PlusLine

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	Administrator
StandardLine - M31R with PlusLine Productivity Pack	

The setup for Reminder Only Test and Standard Test is identical.

To enable a test:

- 1 Navigate to **QA > Test Setup**.
- 2 Select **Reminder Only Test** or **Standard Test**.
- 3 Select **Enabled**.
 - ➔ **Scheduling Mode** becomes active.
- 4 Adjust the settings as required.

Scheduling Settings

- Scheduling Mode:
 - Timed
 - Off
- Due Interval:
 - Adjust the time interval between tests (maximum 24:00 hours).
- Schedule Type:
 - Fixed Schedule - Test scheduled at a fixed rate, regardless of when they are run.
 - Interval From Last - Test scheduled at an interval after the last test was run.
- Overdue After:
 - Set the time period before an overdue warning is given.
 - Deselect the Enabled option to disabled the function.
- Stop When Overdue:
 - When enabled, the conveyor will stop when the overdue warning is given.
 - **Note** That overdue time is capped at half of the due period. For example, due time is 24:00 hours, max overdue is 12:00 hours.

Triggers

- When enabled, the following test triggers will also be included:
 - Test Following Product Change.
 - Test Following Settings Change.
 - Test Following Start-Up.

Access Control

- When enabled, login will be required to undertake the Performance Test.

Note When enabled, Scheduling Mode can be switched to Off, Test Setup will still show as enabled allowing tests to be run without scheduling.

11.3 Carrying out a Performance Test

Depending on settings configured in Test Setup, the conveyor may stop when the test overdue period is exceeded.

- The Start/Stop button will be grayed out until a test Pass is logged.

Prerequisites

- An appropriate test sample as determined by your company's standards for metal detection.
- A sample of the current Product being tested.
 - Multiple product samples may be required, ensure Test Samples are available for each.

Considerations:

- Always pass the test sample through the detector aperture with a sample of the current Product.
- For worst-case detection, pass the test sample through the center of the aperture.
- After every test, make sure that any reject device associated with the metal detector, correctly rejects the test sample.
- Ensure the test sample can be retrieved before passing downstream if the reject device fails.



Important Safety Message

If the Performance Test fails, you must follow your company's procedures and standards according to your metal detection strategy.

When the cause of the failure has been determined and rectified, it may be necessary to retest all products quarantined since the last performance test. This ensures they are free from metal contamination. If re-testing is not possible, it may be necessary to destroy the affected products.

11.3.1 Performance Test - Reminder Only Test

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

Note Undertaking performance tests can be restricted to QA and Administrator only in Test Setup.

The Performance Test (Reminder Only Test) simply logs a test has been reported.

To log a test:

- 1 When a **Performance Test (Reminder Only Test)** is due, undertake the test according to your company's requirements.
- 2 Navigate to **QA > Performance Test**.
- 3 Pass a product, complete with test piece, through the metal detector aperture.
 - ➔ If fitted, ensure the product and test piece are placed upstream of the **Pack Sensor**.
- 4 Monitor the screen, the bargraph will identify if the test piece has been detected.
- 5 Select **Fail** or **Pass**.
- 6 The **Test Due/Test Overdue** warning will only be reset when a test **Pass** is logged.

See also

- 🔗 [Setting up Performance Tests ▶ Page 70](#)
- 🔗 [Managing Alarms ▶ Page 84](#)

11.3.2 Standard Test

Item	Details
PlusLine - M33R, M33RB and M34R	All Users
StandardLine - M31R with PlusLine Productivity Pack	

Note Undertaking performance tests can be restricted to QA and Administrator only in Test Setup.

The Reminder Only Test can be run as well as the Standard Test.

The Standard Test guides the user to perform a series of tests based on the test samples setup for the product under test.

To do this:

- 1 Navigate to **QA > Performance Test**.
- 2 Select **Standard Test**.

- 3 **Start** the conveyor.
- 4 Select the first material to test.
- 5 Pass a product, complete with test piece, through the metal detector aperture.
 - ➔ If fitted, ensure the product and test piece are placed upstream of the **Pack Sensor**.
- 6 Monitor the screen, the following items (where fitted) should show a green tick:
 - ➔ **Pack Sensor**
 - ➔ **Detection**
 - ➔ **Reject Activated**
 - ➔ **Reject Confirmed**
- 7 Red crosses indicate a **Fail**.
 - ➔ Select **Fail**.
 - ➔ Investigate cause and re-run the test.
 - ➔ Check associated Alarms for potential cause.
- 8 If test successful, the **Pass** button turns green.
 - ➔ Select **Pass**.
- 9 Repeat test for each material, as required.
 - ➔ Any failure requires investigation and re-run of the test.
- 10 On successful completion of all material tests, Select **Pass**.

Note

- On successful completion of the required tests, the Performance Monitoring Routine alarms will reset.
- The alarms will not reset until all tests are successful.

See also

-  Test Sample Setup ▶ Page 70
-  Setting up Performance Tests ▶ Page 70
-  Managing Alarms ▶ Page 84

12 Event Logs and Reports

There are a number of reports and logs available to the user which are described in the following section of this manual.

- Logs have a retention period of 90 days before being overwritten.
- Logs have a maximum storage limit of 50,000 individual records.

12.1 Filtering Logs and Reports

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R StandardLine - M31R with PlusLine Protection Pack	QA or Administrator

Note User login is not required to view and filter the following logs:

- Statistics > Detection Log.
- System > > Alarm Log.

The filter  function consists of the following three elements:

Search

Enter a text string into the search box to refine the search result.

Checkbox selection

Additional pages of checkbox options are available, select or deselect options as required to refine the search result.

Note Only options applicable to the selected log will be displayed.

Time frame

The user can enter:

- Start Enabled and End Enabled.
- Start and End Date for filtered results.
- Start and End Time for filtered results.

Note To cancel a filter, select the Filter Cancel icon .

12.1.1 Example - QA > History

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

- 1 Navigate to **QA > History**.
- 2 Select Filter .
- 3 Enter search criteria:
 - ➔ Basic search - enter search string.
 - ➔ Select additional search criteria as required.
 - ➔ Where required, enter **Time** criteria.
- 4 **Accept**.
 - ➔ The search results are presented and a filter cancel icon appears .

12.2 Detection Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R StandardLine - M31R with PlusLine Protection Pack	QA or Administrator

Status indicators are used throughout the report and log pages, in some cases these may be hidden in retracted text blocks.

The following table describes Status Indicator icons used in the detection log:

Status Indicator	Description
	Successful confirmation
	Standard fault preventing operation
	Reject Inhibit
	Reject Actioned

- Navigate to **Statistics > Detection Log**.
 - ➔ The following information is displayed on the **UI**:

Time:

- Column contains date stamped folders (number in brackets indicate number of time stamped records).
- Select cog icon  and tick Group By Product to group product logs.
- Expand folders for individual records.

Event:

- Indicates the type of event:
 - Contaminated Pack = detected contaminant in a discrete product.
 - Contamination = detected contaminant in a batch run of products.
 - Fault Pack = A fault detected in a discrete product (does not have a level of detection value quoted).
 - Fault Rejection = Loose product that has failed inspection due to machine fault.
 - Unsynchronized Contamination = an unexpected detected event (for example, a metallic object falling on the conveyor).
 - Fail Safe Rejection = Pack that could not be inspected by the time it reached the reject, so rejected to be safe.
 - Changeover Pack = Pack inspected during changeover.
 - Changeover = Loose product inspected during changeover
- Expand the event type to display additional information of the selected test.

Details:

- Indicates level of detection (in bars).

12.2.1 Downloading Detection Logs

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R StandardLine - M31R with PlusLine Protection Pack	QA or Administrator

Note Prior to downloading logs, ensure a USB storage device is inserted into the USB port of the system.

- Files will be saved to folder "MDReports\Detections"

A maximum of 10,000 individual logs can be downloaded at any one time.

- 1 Navigate to **QA >> Reports > Detections**.
- 2 Filter results as required.
- 3 Expand and collapse the log results using the  icon.
- 4 Individually select the items to download, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.

- ➔ Tap the icon a second time to include all new and downloaded reports.
 - ➔ Tap the icon a third time to deselect all reports.
- 5 To download, select the green **Download Report** button (only active when a selection has been made).

Note Logs can also be deleted by selecting the red Delete button (only active when a selection has been made).

12.2.2 Deleting the Detection Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

Note When using the delete feature, all logs are deleted, even when records are filtered.

- 1 Navigate to **Statistics > Detection Log**.
- 2 Select the cog icon .
- 3 To delete, select the Clear Detection Log icon.

Note Detection Logs can also be deleted from the Detection Log download page by selecting the red Delete button.

See also

 Downloading Detection Logs ► Page 75

12.3 History

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

Status indicators are used throughout the report and log pages, in some cases these may be hidden in retracted text blocks.

The following table describes Status Indicator icons used in the History log:

Status Indicator	Description
	Passed
	Canceled
	Failed
	Report not downloaded

- Navigate to **QA > History**.
 - ➔ The following information is displayed on the **UI**:

Time:

- Column contains date stamped folders (number in brackets indicate number of time stamped records).
- Expand folders for individual records.

Test:

- Column displays the type of performance test undertaken (expand the time column to view).
- For Standard Tests only, select to blue arrow to display samples tested, their results, and signal strength.

Result:

- Column displays the test result, Passed, Cancelled or Failed.

Details:

- Column displays who conducted the test.

12.3.1 Downloading History Logs

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

Note Prior to downloading logs, ensure a USB storage device is inserted into the USB port of the system.

- Files will be saved to folder "MDReports\TestReports"

A maximum of 5,000 individual sequences can be downloaded at any one time.

- Navigate to **QA > History**.
- Select the Download icon 
- Individually select the items to download, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.
 - ➔ Tap the icon a second time to include all new and downloaded reports.
 - ➔ Tap the icon a third time to deselect all reports.
- To download, select the green **Download Report** button (only active when a selection has been made).

12.3.2 Deleting the History Log

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

Note When using the delete feature, all logs are deleted, even when records are filtered.

- Navigate to **QA > History**.
- Select the Download icon 
- Individually select the items to delete, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.
 - ➔ Tap the icon a second time to include all new and downloaded reports.
 - ➔ Tap the icon a third time to deselect all reports.
- To delete, select the red **Delete** button (only active when a selection has been made).

12.4 Access Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

- Navigate to **System > Users > Access Log**
 - ➔ The following information is displayed on the UI:

Time:

- Column contains date stamped folders (number in brackets indicate number of time stamped records).
- Expand folders for individual records.

Event:

- Displays which user has logged into the UI.
- Expand the user name for a time-line of user activity.

12.4.1 Downloading User Access Logs

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

Note Prior to downloading logs, ensure a USB storage device is inserted into the USB port of the system.

- Files will be saved to folder "MDReports\User Access"

A maximum of 10,000 individual logs can be downloaded at any one time.

- 1 Navigate to **QA >> Reports > User Access**.
- 2 Filter results as required.
- 3 Expand and collapse the log results using the  icon.
- 4 Individually select the items to download, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.
 - ➔ Tap the icon a second time to include all new and downloaded reports.
 - ➔ Tap the icon a third time to deselect all reports.
- 5 To download, select the green **Download Report** button (only active when a selection has been made).

Note Logs can also be deleted by selecting the red Delete button (only active when a selection has been made).

12.4.2 Deleting Access Logs

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

Note When using the delete feature, all logs are deleted, even when records are filtered.

- 1 Navigate to **System > Users > Access Log**
- 2 Select the cog icon .
- 3 To delete, select the red bin .

12.5 Alarms Log

Metal Detector	Access Level Required
PlusLine - M33R, M33RB and M34R	All Users
StandardLine - M31R with PlusLine Protection Pack	

Status indicators are used throughout the report and log pages, in some cases these may be hidden in retracted text blocks.

The following table describes Status Indicator icons used in the alarm log:

Status Indicator	Description
	Alarm acknowledged
	Information level alarm
	Warning level alarm
	Critical fault
	Inactive
	Alarm reset

There are two ways to access Alarm Logs:

- Navigate to System > > Alarm Log
- Navigate to Alarms > 
- The following information is displayed on the **UI**:

Time:

- Column contains date stamped folders (number in brackets indicate number of time stamped records).
- Expand folders for individual records.

Alarm:

- Column contains a description of alarm faults complete with alarm code.

- Select the alarm code to access additional information and guidance.

Action:

- Indicators within the column signify actions taken by the user.

State:

- Indicators within the column display the current state of the alarm condition.

12.5.1 Downloading Alarm Logs

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R StandardLine - M31R with PlusLine Protection Pack	QA or Administrator

Note Prior to downloading logs, ensure a USB storage device is inserted into the USB port of the system.

- Files will be saved to folder "MDReports\Alarms"

A maximum of 10,000 individual logs can be downloaded at any one time.

- 1 Navigate to **QA >> Reports > Alarms**.
- 2 Filter results as required.
- 3 Expand and collapse the log results using the  icon.
- 4 Individually select the items to download, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.
 - ➔ Tap the icon a second time to include all new and downloaded reports.
 - ➔ Tap the icon a third time to deselect all reports.
- 5 To download, select the green **Download Report** button (only active when a selection has been made).

Note Logs can also be deleted by selecting the red Delete button (only active when a selection has been made).

12.5.2 Deleting Alarm Logs

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R StandardLine - M31R with PlusLine Protection Pack	QA or Administrator

Note When using the delete feature, all logs are deleted, even when records are filtered.

Alarm Logs via Alarm Settings.

- 1 Navigate to **System >> Alarm Settings >> Alarm Log**.
- 2 To delete, select the red bin .

Alarm Logs via Alarm Log

- 1 Navigate to **System >> Alarm Log**.
- 2 Select the cog icon .
- 3 To delete, select the red bin .

Alarm Logs via Alarms

- 1 Navigate to **Alarms**.
- 2 Select the cog icon .
- 3 Navigate to **Alarm Log**.
- 4 To delete, select the red bin .

Note You can access the Alarm Log via Alarms without Administrator login, but will not be able to delete records.

12.6 Performance Test

Metal Detector	Access Level Required
All Metal Detectors	QA or Administrator

Status indicators are used throughout the report and log pages, in some cases these may be hidden in retracted text blocks.

The following table describes Status Indicator icons used in the Performance Tests log:

Status Indicator	Description
	Passed
	Canceled
	Failed
	Report not downloaded

- Navigate to **QA > > Reports > Performance Tests**.
 - ➔ The following information is displayed on the **UI**:

Time:

- Column contains date stamped folders (number in brackets indicate number of time stamped records).
- Expand folders for individual records.

Test:

- Column displays the type of performance test undertaken (expand the time column to view).

Details:

- Column displays if the report has been downloaded to a USB storage device.

12.6.1 Downloading the Performance Test

Metal Detector	Access Level Required
PlusLine - M33R, M33RB, M34R	QA or Administrator
StandardLine - M31R with PlusLine Protection Pack	

Note Prior to downloading logs, ensure a USB storage device is inserted into the USB port of the system.

- Files will be saved to folder "MDReports\Test Reports"

A maximum of 5,000 individual sequences can be downloaded at any one time.

- 1 Navigate to **QA > > Reports > Performance Tests**.
- 2 Filter results as required.
- 3 Individually select the items to download, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.
 - ➔ Tap the icon a second time to include all new and downloaded reports.
 - ➔ Tap the icon a third time to deselect all reports.
- 4 To download, select the green **Download Report** button (only active when a selection has been made).

Note Logs can also be deleted by selecting the red Delete button (only active when a selection has been made).

12.6.2 Deleting the Performance Test Log

Note When using the delete feature, all logs are deleted, even when records are filtered.

- 1 Navigate to **QA > > Reports > Performance Tests**.
- 2 Individually select the items to delete, or use the  icon to:
 - ➔ Tap the icon once to select all new reports.

- ➡ Tap the icon a second time to include all new and downloaded reports.
 - ➡ Tap the icon a third time to deselect all reports.
- 3 To delete, select the red **Delete** button (only active when a selection has been made).

13 System Updates

Metal Detector	Access Level Required
All Metal Detectors	Administrator



CAUTION

Ensure power is maintained during system updates

Loss of power during a system update may result in damage to the internal components of the Metal Detector, operating system, or firmware.



NOTICE

Please note that updates may take several hours.

Plan downtime accordingly before proceeding with the procedures described below.

13.1 OS Update

OS Update may be available to your organization, depending on your service contract.

To apply an operating system update:

- 1 Download the ZIP archive containing the **OS Update** file from <http://www.mt.com/pi-patches>.
- 2 Extract the **OS Update** file from the ZIP archive to a folder called **OSUpdate** in the root folder of a USB stick (if the **OSUpdate** folder does not exist, create it).
- 3 Insert the USB stick into the machine where you want the update to take place.
- 4 Navigate to **System > > Self Service**.
- 5 The **Challenge Login** page opens, asking you for a response code. To obtain the response code, take a note of the serial number and challenge code displayed on the page and contact your METTLER TOLEDO representative.
- 6 Once you have the response code, select **Enter Response Code**.
- 7 Enter the response code.
- 8 **Accept**.
- 9 On the next page, select **OS Update**.
- 10 On the **OS Update** page, select the update file from your USB stick and follow the screen prompts.

13.2 System Update

System Update may be available to your organization, depending on your service contract.

Note Firmware can be upgraded or downgraded using this function.

To apply a system update:

- 1 Download the ZIP archive containing the **System Update** file from <http://www.mt.com/pi-patches>.
- 2 Extract the **System Update** file from the ZIP archive to a folder called **mdsoftware** in the root folder of a USB stick (if the **mdsoftware** folder does not exist, create it).
- 3 Insert the USB stick into the machine where you want the update to take place.
- 4 Navigate to **System > > Self Service**.
- 5 The **Challenge Login** page opens, asking you for a response code. To obtain the response code, take a note of the serial number and challenge code displayed on the page and contact your METTLER TOLEDO representative.
- 6 Once you have the response code, select **Enter Response Code**.
- 7 Enter the response code.
- 8 **Accept**.
- 9 On the next page, select **System Update**.

10 On the **System Update** page, select the update file from your USB stick and follow the screen prompts.

13.3 IO Expanders

An optional IO Expansion Board is available, this board increases available system inputs X6 and outputs X6. When an expansion board is installed, it is automatically detected by the system. The additional inputs and outputs can be configured from System > > System Integration.

If at any time the board is removed or replaced, it must be deleted from the system to prevent system alarms.

To remove the IO Expansion Board from the UI:

- 1 Navigate to **System > > Self Service**.
- 2 The **Challenge Login** page opens, asking you for a response code. To obtain the response code, take a note of the serial number and challenge code displayed on the page and contact your METTLER TOLEDO representative.
- 3 Once you have the response code, select **Enter Response Code**.
- 4 Enter the response code.
- 5 **Accept**.
- 6 On the displayed screen, select IO Expanders.
- 7 Select the IO Expansion Board to be removed.
- 8 Select Remove.
 - ➔ The IO Expansion Board is removed from the system.

14 Managing Alarms

Metal Detector	Access Level Required
All Metal Detectors	Depends on settings. See below

Alarms are a way for the metal detector to alert the user that certain system conditions are present. Some alarms also initiate a response from the system. For example, an alarm activated by a potentially dangerous situation may initiate a loud audible signal (the user is alerted) and pause the production line until the problem is resolved (the system responds to the danger).

Each alarm has a severity level, which could be low, medium, high or very high.

Alarms often require the user to respond. There are two types of response - reset and acknowledgment - they can both be required by the same alarm.

Some alarms can be reset automatically, as the Trigger is removed.

An alarm could also be set up to latch, that is to remain active until it is reset manually by the user, even after the Trigger is removed.

In addition, some alarms require action from the user as acknowledgment, that is before the alarm can be reset. Both acknowledging and resetting a latched alarm can change the behavior of the system. For example, acknowledging an alarm may stop the alarm sound but the production line could remain paused until the alarm is reset manually.

See also

- [Description of the User Interface \(UI\) ▶ Page 16](#)
- [Choosing Alarm Settings ▶ Page 84](#)
- [Choosing the Display Priority ▶ Page 85](#)
- [Choosing how Alarms are Acknowledged ▶ Page 85](#)
- [Choosing how Alarms are Reset ▶ Page 85](#)
- [Acknowledging Alarms ▶ Page 86](#)
- [List of Alarms ▶ Page 86](#)

14.1 Alarm Icons and their Meaning

The Alarms icon informs you of the present running condition of the metal detector.

The five possible states are:



System Healthy.

Alarms



The metal detector is working, but you are notified that there is a condition to be aware of. The number in the red circle indicates the number of notifications. Pressing the icon will list the notifications.

Alarms



The metal detector needs attention. A fault could develop. The number in the red circle indicates the number of notifications. Pressing the icon will list the notifications.

Alarms



The metal detector has stopped working. The number in the red circle indicates the number of notifications. Pressing the icon will list the notifications.

Alarms



The metal detector has developed a critical fault. This appears for faults related to safety issues, such as Safety Device Operated.

Alarms

14.2 Choosing Alarm Settings

Metal Detector	Access Level Required
All Metal Detectors	QA, Engineer or Administrator

- 1 Navigate to **System > > Alarm Settings**.
- 2 In **Alarm Settings**, you can:
 - ➔ Choose the **Display Priority**, which determines which alarm message is the most prominent message on the **UI**. The most prominent message appears on the **Home** page.
 - ➔ Choose how the alarms are acknowledged and reset.

14.2.1 Choosing the Display Priority

Metal Detector	Access Level Required
All Metal Detectors	QA, Engineer or Administrator

- 1 Navigate to **System > > Alarm Settings**.
- 2 Select the **Display Priority**:
 - ➔ **Highest Severity** causes the most severe alarm to appear on the **Home** page, if there are multiple alarms. The list on the **Alarms** page will also be ordered by severity.
 - ➔ **First Out** causes the earliest active alarm to appear on the **Home** page, if there are multiple alarms.
 - ➔ **Last Out** causes the latest active alarm to appear on the **Home** page, if there are multiple alarms.

See also

[Managing Alarms](#) ▶ Page 84

14.2.2 Choosing how Alarms are Acknowledged

Metal Detector	Access Level Required
All Metal Detectors	Full Administrator Partial QA, Engineer

Note QA and Engineer have access only to Acknowledge Inactive Alarms Automatically.

- 1 Navigate to **System > > Alarm Settings**.
- 2 Select **Local Acknowledge Access**:
 - ➔ **Enabled** - users can acknowledge the alarm via the UI.
 - ➔ **Access Controlled** - users are required to be logged in to acknowledge the alarm via the UI.
 - ➔ **Disabled** - the alarm can only be acknowledged by an input signal.
- 3 The checkbox **Acknowledge Inactive Alarms Automatically** allows you to choose automatic acknowledgment of alarms that are no longer active.

See also

[Managing Alarms](#) ▶ Page 84

14.2.3 Choosing how Alarms are Reset

Metal Detector	Access Level Required
All Metal Detectors	QA, Engineer or Administrator

- 1 Navigate to **System > > Alarm Settings**.
- 2 Select **Local Reset Access**:
 - ➔ **Enabled** - users can reset the alarm via the **UI**.
 - ➔ **Access Controlled** - users are required to be logged in to reset the alarm via the **UI**.
 - ➔ **Disabled** - the alarm can only be reset by an input signal.

See also

[Managing Alarms](#) ▶ Page 84

14.3 Acknowledging Alarms

Metal Detector	Access Level Required
All Metal Detectors	Depends on settings. See below

Note The administrator sets who has acknowledging rights.

Alarms are acknowledged manually or automatically, depending on the type of alarm, and depending on how the system is set up. When an alarm is acknowledged, a tick appears in the checkbox on the Acknowledge icon.

To acknowledge an individual alarm manually:

- 1 Press the **Alarms** icon.
- 2 Then select **Acknowledge**.

To acknowledge or reset all alarms, use the toolbar at the top of the Alarms page.

- The tick acknowledges all alarms.
- The circular arrow resets all alarms, with the exception of overdue performance test alarms.
- **Result:** The alarm icon on the Alarms page will indicate that the alarm is Acknowledged.

See also

- 🔗 Managing Inputs ► Page 41
- 🔗 Managing Alarms ► Page 84
- 🔗 Choosing the Display Priority ► Page 85
- 🔗 Choosing how Alarms are Acknowledged ► Page 85
- 🔗 Choosing how Alarms are Reset ► Page 85

14.4 List of Alarms

Code	Name	Guidance	Severity	Category
2000	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2001	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2002	Event Scheduling Failed	Remove all product from system and re-run. Reduce product flow rate to prevent reoccurrence	High (Fault)	Abort
2003	Event Scheduling Failed	Remove all product from system and re-run. Reduce product flow rate to prevent reoccurrence	High (Fault)	Abort
2004	Internal Failure	Contact METTLER TOLEDO Service	High (Fault)	Abort
2005	Pack Rate Too High	Remove product at Pack Sensor and pass again. Increase pack spacing to prevent reoccurrence	High (Fault)	Abort
2006	Reject Operation Not Confirmed	Identify and remove contaminated product	High (Fault)	Stop
2007	Rejected Product Not Confirmed	Identify and remove contaminated product	High (Fault)	Stop
2008	Detection Event	Check Pack Sensor. Identify and remove contamination	High (Fault)	Stop
2009	Conveyor Running Fault	Check speed sensor	High (Fault)	Abort
2010	Speed Sensor Fault	Check speed sensor	High (Fault)	Abort
2011	Battery Low	Contact METTLER TOLEDO Service	Medium (Warning)	No Action
2012	Battery Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2013	Conveyor Running Not Confirmed	Check conveyor running input	High (Fault)	Stop

Code	Name	Guidance	Severity	Category
2014	Conveyor Stopped	Start conveyor through external system. If alarm persists, check 'conveyor running' feedback signal.	Low (Info)	Stop
2015	Internal Failure	Contact METTLER TOLEDO Service if alarm reoccurs	High (Fault)	Abort
2016	Internal Failure	Power cycle to clear. Contact METTLER TOLEDO Service if alarm reoccurs	High (Fault)	Abort
2017	Internal Failure	Contact METTLER TOLEDO Service	High (Fault)	Abort
2018	Pack Sensor Blocked	Clear blockage. Remove all product from the system and re-run	High (Fault)	Stop
2019	Power Supply Fault	Check Input and Output Connections	High (Fault)	Abort
2020	Power Supply Fault	Check Input and Output Connections	High (Fault)	Abort
2021	Fan Fault	Contact METTLER TOLEDO Service	Medium (Warning)	No Action
2022	Safety Device Operated	Make safe and reset safety circuit	Very High (Critical)	Abort (E-Stop)
2023	Detection Event	Remove contaminated product from conveyor	Medium (Warning)	Stop
2024	Downstream Blocked	Resolve downstream problem	Medium (Warning)	Stop
2025	Upstream Starved	Resolve upstream problem	Medium (Warning)	Stop
2026	Start Input Not Configured	Configure input to 'Start' function	High (Fault)	Stop
2027	Stop Input Not Configured	Configure input to 'Stop' function	High (Fault)	Stop
2028	Conveyor Drive Output Not Configured	Configure output to 'Conveyor Drive' function	High (Fault)	Stop
2029	Safety Circuit Input Not Configured	Configure input to 'Safety Circuit' function	High (Fault)	Stop
2030	Conveyor Running Input Not Configured	Configure an input to 'Conveyor Running' function	High (Fault)	Stop
2031	System Stop Output Not Configured	Configure an output to 'System Stop' function	High (Fault)	Stop
2032	Clearing Detection Event	Run belt until manually cleared section is downstream	Low (Info)	No Action
2033	Service Required	Contact METTLER TOLEDO Service to service the device	Low (Info)	No Action
2034	CPU Fault	Power cycle to clear. Contact METTLER TOLEDO Service if alarm reoccurs	Medium (Warning)	Stop
2035	Developer Build	Contact METTLER TOLEDO Service	Low (Info)	No Action
2036	Fan Fault	Contact METTLER TOLEDO Service	Medium (Warning)	No Action
2037	Reject Inhibited	Disable the reject inhibit option to return to normal production mode.	Low (Info)	No Action
2038	Event Scheduling Failed	Remove all product from system and re-run. Reduce product flow rate to prevent reoccurrence	High (Fault)	Abort
2039	Detection Event	Remove contaminated product from conveyor	Low (Info)	No Action
2040	Rejected Product Sensor Fault	Check that nothing is obstructing the sensor, and that the input is configured correctly.	High (Fault)	Abort

Code	Name	Guidance	Severity	Category
2041	Reject Operation Sensor Fault	Check that nothing is obstructing the sensor, and that the input is configured correctly.	High (Fault)	Abort
2042	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2043	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2044	Fail Safe Rejection	Increase the 'Pack to Reject' product setting.	Medium (Warning)	No Action
2045	Bin Full	Empty the reject bin	High (Fault)	Stop
2046	Bin Full	Empty the reject bin	Medium (Warning)	No Action
2047	Air Supply Failure	Replenish Air Supply	High (Fault)	Stop
2048	Air Supply Failure	Replenish Air Supply	Medium (Warning)	No Action
2049	Fail Safe Stop	Increase the 'Pack to Alarm' product setting.	Medium (Warning)	Stop
2050	Contaminated Pack Not Rejected	Please check reject is functioning correctly.	High (Fault)	Stop
2051	Reject Check Sensor Blocked	Please check whether Reject Check Sensor is blocked	High (Fault)	Stop
2052	Pack(s) Missing at Reject Check Sensor	Please check for obstruction and check reject settings	High (Fault)	Stop
2053	Unexpected Pack(s) at Reject Check Sensor	Unexpected pack(s) seen on the belt.	High (Fault)	Stop
2054	Pack Space Warning	Check pack space monitoring	Medium (Warning)	No Action
2055	Reject Guard Fault	Check pack space monitoring	High (Fault)	Stop
2056	Reject Bin Unlocked	Close the reject bin	Medium (Warning)	No Action
2057	Reject Bin Unlocked	Close the reject bin	High (Fault)	Stop
2400	Signal Balance Warning	Contact METTLER TOLEDO Service for a service visit	Medium (Warning)	No Action
2401	Signal Balance Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2402	Power Drive Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2403	Power Supply Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2404	Self Check Phase Error	Contact METTLER TOLEDO Service	High (Fault)	Abort
2405	Self Check Magnitude Error	Contact METTLER TOLEDO Service	High (Fault)	Abort
2406	Power Drive Warning	Contact METTLER TOLEDO Service for a service visit	Medium (Warning)	No Action
2407	Self Check Phase Warning	Contact METTLER TOLEDO Service for a service visit	Medium (Warning)	No Action
2408	Self Check Magnitude Warning	Contact METTLER TOLEDO Service for a service visit	Medium (Warning)	No Action
2409	Fan Fault	Contact METTLER TOLEDO Service	Medium (Warning)	No Action
2410	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2411	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2412	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2413	Developer Build	Contact METTLER TOLEDO Service	Low (Info)	No Action
2414	System Speed Invalid	Update speed settings to be within the usable range of the device.	High (Fault)	Abort
2415	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort

Code	Name	Guidance	Severity	Category
2416	CPU Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2417	CPU Fault	Contact METTLER TOLEDO Service	Low (Info)	No Action
2418	Signal Balance Warning	Clear aperture of any large metal or product. Contact METTLER TOLEDO Service if alarm remains active.	Medium (Warning)	No Action
2419	CPU Fault	If alarm cannot be cleared please restart the machine. If problem persists contact METTLER TOLEDO Service.	High (Fault)	Abort
2420	Signal Balance Warning	Contact METTLER TOLEDO Service for a service visit	Medium (Warning)	No Action
2421	Signal Balance Fault	Contact METTLER TOLEDO Service	High (Fault)	Abort
2422	Balance Recovery Disabled	Power cycle to clear. Contact METTLER TOLEDO Service if alarm reoccurs	Low (Info)	No Action
2423	Signal Balance Fault	Clear aperture of any large metal or product. Contact METTLER TOLEDO Service if alarm remains active.	High (Fault)	Abort
2424	Balance Recovery Active	This alarm should clear automatically. If it does not, contact METTLER TOLEDO Service.	Medium (Warning)	No Action
2500	Internal Failure	Power cycle to clear. Contact METTLER TOLEDO Service if alarm reoccurs	High (Fault)	Abort
2501	Initialization Error	Do a firmware update	High (Fault)	Abort
2502	Initialization Error	Do a firmware update	High (Fault)	Abort
2503	Product Not Setup	Run Product Setup Wizard	Medium (Warning)	Stop
2504	Frequency Calibration Error	Do a frequency calibration	High (Fault)	Abort
2505	Date/Time Not Set	Set the date and time	Low (Info)	No Action
2506	RTC Fault	Contact METTLER TOLEDO Service	Medium (Warning)	No Action
2508	Performance Test Due	An authorized QA user should run a performance test.	Medium (Warning)	No Action
2509	Product Needs Auto Setup	Do an Auto Setup	Medium (Warning)	Stop
2518	Internal Failure	The system has automatically recovered from a fault. Contact METTLER TOLEDO Service.	Very High (Critical)	Abort
2519	Fail Safe Detect	Caused by active settings change, or another alarm	Low (Info)	No Action
2520	Internal Failure	The system has automatically recovered from a fault. Contact METTLER TOLEDO Service.	Very High (Critical)	Abort
2521	Update Failed	An update failed to apply and the system has recovered automatically. Contact METTLER TOLEDO Service.	High (Fault)	Abort
2523	Internal Failure	The system has automatically recovered from a fault. Contact METTLER TOLEDO Service.	High (Fault)	Abort
2524	Internal Failure	The system has automatically recovered from a fault. Contact METTLER TOLEDO Service.	High (Fault)	Abort

Code	Name	Guidance	Severity	Category
2525	Internal Failure	The system has automatically recovered from a fault. Contact METTLER TOLEDO Service.	High (Fault)	Abort
2526	OS Not Activated	Contact METTLER TOLEDO Service	Low (Info)	No Action
2527	Write Filter Not Enabled	Contact METTLER TOLEDO Service	Low (Info)	No Action
2528	Developer Mode	Remove the developer authentication certificate	Low (Info)	No Action
2529	SD Card Failure	Check SD card is properly inserted. Contact METTLER TOLEDO Service.	High (Fault)	No Action
2530	Reject Check Not Setup	Run Auto Reject Check Calibration	Medium (Warning)	Stop
2531	ProdX Server Unavailable	Prodx Server connection has been lost, please check network connectivity	High (Fault)	Stop
2532	ProdX Server Unavailable	Prodx Server connection has been lost, please check network connectivity	Medium (Warning)	No Action
2533	Performance Test Overdue	An authorized QA user should run a performance test.	High (Fault)	Stop
2534	Performance Test Overdue	An authorized QA user should run a performance test.	High (Fault)	No Action
2535	Performance Test Timing Lost	Setup performance tests.	High (Fault)	No Action
2536	Performance Test Store Almost Full	An authorized QA user should delete test reports	Low (Info)	No Action
2537	Performance Test Store Full	Reports will be automatically deleted. An authorized QA user should delete test reports	Medium (Warning)	No Action
2538	Alarm Log Store Almost Full	Old events will be automatically deleted	Low (Info)	No Action
2539	Alarm Log Store Full	Old events will be automatically deleted	Medium (Warning)	No Action
2540	Detection Log Store Almost Full	Old events will be automatically deleted	Low (Info)	No Action
2541	Detection Log Store Full	Old events will be automatically deleted	Medium (Warning)	No Action
2542	Access Log Store Almost Full	Old events will be automatically deleted	Low (Info)	No Action
2543	Access Log Store Full	Old events will be automatically deleted	Medium (Warning)	No Action
2544	Demonstration Firmware Version	The running firmware is intended for demonstration only. Contact METTLER TOLEDO Service.	Low (Info)	No Action
2545	Invalid Data Detected	Some settings have been reset. Confirm that your settings are correct before proceeding.	Very High (Critical)	Abort
2546	Expander IO Unassigned	IO functions have been unassigned from expansion board and set to unused. Check IO functions.	Very High (Critical)	Abort
2547	Unassigned IO Reset	Unassigned IO functions have been set to unused. Check IO functions.	Very High (Critical)	Abort
2548	Developer Build	Contact METTLER TOLEDO Service	Low (Info)	No Action
2549	Developer Build	Contact METTLER TOLEDO Service	Low (Info)	No Action

15 Preventative Maintenance

It is the users responsibility to ensure the metal detector and associated equipment is cleaned, inspected and maintained on a regular basis. METTLER TOLEDO recommend that the user develops a cleaning and maintenance schedule that reflects the operating conditions in which the metal detector is placed.

METTLER TOLEDO Service ► www.mt.com/slmd-contact are happy to provide support when planning your maintenance schedule and service requirements.

The following schedule is for guidance only and provides the minimum maintenance requirements, based on normal operating conditions:

	Daily	Weekly	Monthly	Annually
Clean and visually inspect metal detector for signs of damage	✓			
Performance testing (of same product run)*	✓			
Visually inspect metal detector liner for damage, cracks or sagging				✓
Visually inspect head cable for damage				✓
Check cable glands are tightened correctly				✓
Visually inspect the HMI seal. Check it is mounted correctly and securing bolts are correctly torqued				✓
Check the PSU enclosure for damage				✓
Ensure PSU enclosure front panel is mounted correctly and securing bolts correctly torqued				✓
Check head and PSU enclosure mounting bolts are correctly torqued				✓

* Frequency based on your company's standards and quality assurance procedures.

* Must also be undertaken after a product change or any system/settings change.

Torque settings:

- HMI securing bolts = 5 N/m
- PSU enclosure bolts = 5 N/m

16 Technical Specifications

16.1 Metal Detectors

M31R StandardLine	Entry-level solution offers versatile, reliable inspection for unpacked or packaged applications, with smart, digital SENSE technology inside
M33R PlusLine	Delivers enhanced performance for dry products, with smart, digital SENSE technology inside
M34RB PlusLine	Delivers enhanced performance for bulk products, with smart, digital SENSE technology inside
M34R PlusLine	Delivers enhanced performance for wet (conductive) products, with smart, digital SENSE technology inside

16.2 Feature Packs

Productivity Pack (PlusLine)	PMR Wizard Additional user accounts Additional product memories
Protection Pack (PlusLine)	Due diligence logs and reports Bin lock/unlock Reject Check Alarm Log Access Log Detection Log

To protect your product's future:

METTLER TOLEDO Service assures the quality, measuring accuracy and preservation of value of this product for years to come.

Please request full details about our attractive terms of service.

► www.mt.com/service

www.mt.com

For more information

Mettler-Toledo Safeline Ltd.

Montford Street, Salford
M50 2XD, England, UK

For technical support, please contact your local METTLER TOLEDO Safeline service representative:
www.mt.com/contact

Subject to technical changes.

© 11/2023 METTLER TOLEDO. All rights reserved.
30710293 C en



30710293